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## * Multiple Correlation Coefficient

→ It is the study of relationship between one dependent and combined effect of two or more than two independent variables.

let us consider, three variables $x_1, x_2$ & $x_3$. Then,

- $R_{1.23}$ = Multiple correlation coefficient between $x_1$ & combined effect of $x_2$ & $x_3$

- $R_{2.13}$ = Multiple correlation coefficient between $x_2$ & combined effect of $x_1$ & $x_3$

- $R_{3.12}$ = Multiple correlation coefficient betn $x_3$ & combined effect $x_1$ & $x_2$.

## ⊛ Computational Formulae

- $R_{1.23} = \sqrt{\dfrac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$

- $R_{2.13} = \sqrt{\dfrac{r_{12}^2 + r_{23}^2 - 2r_{12}r_{23}r_{13}}{1 - r_{13}^2}}$

- $R_{3.12} = \sqrt{\dfrac{r_{13}^2 + r_{23}^2 - 2r_{13}r_{12}r_{23}}{1 - r_{12}^2}}$

Note :-

(i) Value of multiple Correlation coefficient lies between 0 & 1.

(ii) Multiple correlation coefficient is symmetrical.

i.e. $R_{i \cdot jk} = R_{i \cdot kj}$

(*) Partial Correlation Coefficient

→ It is the correlation between one dependent and one independent keeping the effect of all remmaining variable as const.

let us consider three variable $X_1, X_2$ & $X_3$. Then,

- $X_{12 \cdot 3}$ = Partial Correlation Coefficient between $X_1$ & $X_2$ keeping $X_3$ as const.

- $X_{13 \cdot 2}$ = Partial Correlation Coefficient between $X_1$ & $X_3$ keeping $X_2$ as const.

- $X_{23 \cdot 1}$ = Partial Correlation Coefficient between $X_2$ & $X_3$ keeping $X_1$ as const.

Computational Formulae

- $$r_{12 \cdot 3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2}\sqrt{1 - r_{23}^2}}$$

- $$r_{13\cdot 2} = \frac{r_{13} - r_{12}\, r_{23}}{\sqrt{1 - r_{12}^2}\sqrt{1 - r_{23}^2}}$$

- $$r_{23\cdot 1} = \frac{r_{23} - r_{12}\, r_{13}}{\sqrt{1 - r_{12}^2}\sqrt{1 - r_{13}^2}}$$

**Note:-**

(1) Partial correlation coefficient lies between $-1$ to $1$

(2) It is symmetrical ie $r_{ij\cdot k} = r_{ji\cdot k}$.

## (*) Coefficient of Multiple determination

→ It is the square of multiple correlation coefft. & represents the percentage of total variation of dependent variables due to variation of independent variable used in the Model.

E.g If $R = 0.9$ then coefft. of multiple determination $R^2 = 0.9^2 = 0.81$ or $81\%$ of total ~~defi~~ variation of dependent variable is due to variation of independent variable used in the study.

## (*) Computational Formulae

* $$R^2 = \frac{a\Sigma y + b_1 \Sigma X_1 Y + b_2 \Sigma X_2 Y - n\bar{y}^2}{\Sigma y^2 - n\bar{y}^2}$$

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or,

* $R^2 = \frac{SSR}{TSS}$

where $SSR$ = Sum of square due to regn
$= \sum(\hat{y} - \bar{y})^2$

$TSS = \sum(y-\bar{y})^2$ = Total sum of square

$SSE = \sum(y-\hat{y})^2$ = Sum of square due to Error

Here, $TSS = SSR + SSE$

## * Standard Error of Estimate

→ It is the S.D. of estimated value from regression equation. It is computed as.

* $$S_{yx} = \sqrt{\frac{\sum y^2 - a\sum y - b_1\sum y x_1 - b_2 \sum y x_2}{n-k-1}}$$

or,

$$S_{yx} = \sqrt{MSSE}$$

where,

$n$ = No. of data

$k$ = no. of independent variable

$MSSE$ = Mean Sum of square due to Error

$= \frac{SSE}{n-k-1}$

## Multiple Regression Equation

→ It is the mathematical relationship betn one dependent & two or more than two independent variables. Let us conside three variables $y, x_1$ & $x_2$ then, Multiple regression eqn of $y$ on $x_1$ & $x_2$ is written as,

$$y = a + b_1 x_1 + b_2 x_2 \quad \text{--- (i)}$$

where,

$y$ = Dependent variable
$x_1$ = 1st independent variable
$x_2$ = 2nd independent variable
$a$ = y-intercept
$b_1$ = Regn coefft. of $y$ on $x_1$ or
Rate of change in $y$ w.r.t $x_1$
$b_2$ = Regn coefft. of $y$ on $x_2$ or
Rate of change in $y$ w.r.t $x_2$.

Normal eqns to solve for $a$, $b_1$ & $b_2$ ar

$$\Sigma y = na + b_1 \Sigma x_1 + b_2 \Sigma x_2 \quad \text{--- (ii)}$$

$$\Sigma y x_1 = a \Sigma x_1 + b_1 \Sigma x_1^2 + b_2 \Sigma x_1 x_2 \quad \text{--- (iii)}$$

$$\Sigma y x_2 = a \Sigma x_2 + b_2 \Sigma x_2^2 + b_1 \Sigma x_2 x_1 \quad \text{--- (iv)}$$

we use Cramer's rule to solve these normal equations.

## Hypothesis Testing

→ It is the process of testing the assumption about significance of relationship between parameter & specified value. There are 2 types of hypothesis

1) Null Hypothesis
2) Alternative Hypothesis

1) Null Hypothesis :-

It is the assumption of no significant difference betn statistics & parameter.

It is denoted by $H_0$.

Eg: $H_0: \mu = 70$ i.e. There is no significant difference in popn mean & its specified value 70.

2) Alternative Hypothesis :-

It is the assumption about statistics & parameter other than Null Hypothesis. It is denoted by $H_1$.

Eg: $H_1: \mu = 50$, Alternative hypothesis maybe

vs, $H_1: \mu \neq 50$ (Two Tailed Test)

or, $H_1: \mu > 50$ (Right Tailed Test)

or, $H_1: \mu < 50$ (Left Tailed Test)

## ⊛ Procedure in Hypothesis Testing

1. Setting of Hypothesis
2. Selection of test- statistic (cal. value)
3. Level of significance ($\alpha$)
4. Table or critical value (tab. value)
5. Decision Making

### Decision Making Procedure

# Table or Critical value approach:-

~~If cal.~~

1.) If cal. value $\leq$ Tab. value we accept $H_0$ & reject

2.) If cal. value > Tab. value we reject $H_0$ & accept $H_1$

# P-value approach:-

1.) If P- value $\leq \alpha$, we reject $H_0$ & accept $H_1$

2.) If P_value > $\alpha$, we accept $H_0$ & reject $H_1$

### P- value

⊛ For one tailed test

$$P\text{- value} = p(z > |z_{cal.}|)$$

⊛ For two- tailed test

$$P\text{-value} = 2 \times p(z > |z_{cal.}|)$$

$\alpha$ = level of significance or risk (Generally we should take $\alpha = 5\%$)

(*) Test of significance of single independent variable

Test - statistic

$$t = \frac{b_i}{Se_i}$$

(*) Test of overall significance of Model

Test - statistics

$$F = \frac{MSSR}{MSSE}$$

where,

MSSR = mean sum of square due to $reg^n$

$= \frac{SSR}{d.f}$

MSSE = mean sum of square due to error

$= \frac{SSE}{df.}$

df = degree of freedom

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(Q.9) Here we have,

$$r_{12} = 0.5,\ r_{23} = 0.1,\ r_{13} = 0.4$$

For $r_{12.3}$

$$r_{12.3} = \frac{r_{12} - r_{13} \cdot r_{23}}{\sqrt{1 - r_{13}^2}\sqrt{1 - r_{23}^2}}$$

$$= \frac{0.5 - 0.4 \times 0.1}{\sqrt{1 - 0.4^2}\sqrt{1 - 0.1^2}}$$

$$= \cancel{0.419}\ 0.504$$

For $r_{13.2}$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{23}}{\sqrt{1 - r_{12}^2}\sqrt{1 - r_{23}^2}}$$

$$= \frac{0.4 - 0.1 \cdot 0.5}{\sqrt{1 - 0.5^2}\sqrt{1 - 0.1^2}}$$

$$= 0.406$$

Hence,

$$\left.\begin{aligned} r_{12.3} &= 0.504 \\ r_{13.2} &= 0.406 \end{aligned}\right\}$$

10)

$r_{12} = 0.4$, $r_{23} = 0.5$, $r_{13} = 0.6$

i) $R_{1.23}$ ii) $R_{23.1}$ iii) $R_{1.23}^2$ iv) $R_{23.1}^2$

i) For $R_{1.23}$

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2 r_{12} r_{23} r_{13}}{1 - r_{23}^2}}$$

$$= \sqrt{\frac{(0.4)^2 + (0.6)^2 - 2\times 0.4\times 0.5\times 0.6}{1-(0.5)^2}}$$

$$= 0.611$$

ii) For $R_{23.1}$

$$R_{23.1} = \frac{r_{23} - r_{12} r_{23}}{\sqrt{1-r_{12}^2}\sqrt{1-r_{13}^2}}$$

$$= \frac{0.5 - 0.6\times 0.4}{\sqrt{1-(0.6)^2}\sqrt{1-(0.4)^2}}$$

$$= 0.35$$

iii) $R_{1.23} = (R_{1.23})^2$

$= 0.611$

$= 0.33$

$R^2 = 0.33 = 33\%$ of total variation of dependent variable is due to variation of independent variable used in the study.

(ii) $R_{23\cdot 1}^2 = (R_{23\cdot 1})^2$

$= \cancel{0.611}\ (0.35)^2$

$= 0.1225$

Q.11

To test ~~pers~~ consistency of data, we should compute higher order

$$r_{13\cdot 2} = \frac{r_{13} - r_{12} r_{23}}{\sqrt{1-r_{12}^2}\sqrt{1-r_{23}^2}}$$

$$= \frac{-0.65 - 0.6\times 0.8}{\sqrt{1-(0.8)^2}\sqrt{1-(0.6)^2}}$$

$$= \cancel{0.504} - 2.041\cancel{47}$$

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Given

$r_{12} = 0.77,\ r_{23} = 0.52,\ r_{13} = 0.72$

(i)
$$r_{12\cdot 3} = \frac{r_{12} - r_{23} r_{13}}{\sqrt{1-r_{23}^2}\sqrt{1-r_{13}^2}}$$

$$= \frac{0.77 - 0.52\times 0.72}{\sqrt{1-(0.52)^2}\sqrt{1-0.72^2}}$$

$= 0.66$

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(22) $r_{13.2} = \dfrac{r_{13} - r_{23}\, r_{12}}{\sqrt{1-(r_{23})^2}\,\sqrt{1-(r_{12})^2}}$

$= \dfrac{0.72 - 0.52 \times 0.77}{\sqrt{1-(0.52)^2}\,\sqrt{1-(0.77)^2}}$

$= 0.58$

(23) $R_{23.1} = \dfrac{r_{23} - r_{12}\, r_{13}}{\sqrt{1-(r_{12})^2}\,\sqrt{1-(r_{13})^2}}$

$= \dfrac{0.52 - 0.77 \times 0.72}{\sqrt{1-(0.77)^2}\,\sqrt{1-(0.72)^2}}$

$= -0.077$

Q. 15) Here we have,

$n = 10$, $\sum x_1 = 10$, $\sum x_2 = 20$, $\sum x_3 = 30$, $\sum x_1^2 = 20$

$\sum x_2^2 = 68$, $\sum x_3^2 = 170$, $\sum x_1 x_2 = 10$, $\sum x_1 x_3 = 15$

$\sum x_2 x_3 = 64$

First, we should compute simple correln. coeff.

Here,

$$r_{12} = \frac{n \sum x_1 x_2 - \sum x_1 \sum x_2}{\sqrt{n \sum x_1^2 - (\sum x_1)^2}\sqrt{n \sum x_2^2 - (\sum x_2)^2}}$$

$$r_{13} = \frac{n \sum x_1 x_3 - \sum x_1 \cdot \sum x_3}{\sqrt{n \sum x_1^2 - (\sum x_1)^2}\sqrt{n \sum x_3^2 - (\sum x_3)^2}}$$

$$r_{23} = \frac{n \sum x_2 x_3 - \sum x_2 \sum x_3}{\sqrt{n \sum x_2^2 - (\sum x_2)^2}\sqrt{n \sum x_3^2 - (\sum x_3)^2}}$$

Now, solving

$$r_{12} = \frac{10 \times 10 - 10 \times 20}{\sqrt{10 \times 20 - 10^2}\sqrt{10 \times 68 - 20^2}} = -0.598 \quad \cancel{-0.007}$$

$$r_{13} = \frac{10 \times 15 - 10 \times 30}{\sqrt{10 \times 20 - 10^2}\sqrt{10 \times 170 - 30^2}} = -0.53$$

$$r_{23} = \frac{10 \times 64 - 20 \times 30}{\sqrt{10 \times 68 - 20^2}\sqrt{10 \times 170 - 30^2}} = 0.0845$$

$y$ = observed value
$\hat{y}$ = Estimated value

* For $r_{12.3}$

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2}\sqrt{1 - r_{23}^2}}$$

$$= \frac{-0.598 - (-0.53 \times 0.084)}{\sqrt{1-(-0.53)^2}\sqrt{1-(0.084)^2}}$$

$$= -0.655$$

* for $R_{1.23}$

$$r_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2 r_{12} r_{23} r_{13}}{1 - r_{23}^2}}$$

$$= \sqrt{\frac{(-0.598)^2 + (-0.53)^2 - 2\times(-0.598)(0.084)(-0.053)}{1-(0.084)^2}}$$

$$= 0.76$$

Q17. Here we have
$n = 20$, $b_1 = 4$, $b_2 = 3$, $Sb_1 = 1.2$, $Sb_2 = 0.8$.

For the test of significance of 1st independent variable

Here we set up hypothesis as

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$H_0: B_1 = 0$ ie first independent variable is insignificant

vs $H_1: B_1 \neq 0$ (TTT) ie first independent variable is significant

Under $H_0$, test static

$$t = \frac{b_1}{s\,b_1}$$

$$= \frac{4}{1.2}$$

$$= 3.333$$

Here,

* level of sig. $(\alpha) = 0.05$

d.f. $= n-k-1 = 20-2-1$

$= 17$

* From t - table

$t_{tab} = 2.110$

* Decision:-

Since $t_{cal.} > t_{tab.}$; we reject $H_0$ & accept $H_1$ with the conclusion that first independent variable is significant.

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For the test of significance of 2nd independent variable

Here we get up hypothesis as

$H_0 : \beta_2 = 0$ ie second independent variable is insignificant

vs $H_1 : \beta_2 \neq 0$ (TTT) ie second independent variable is significant

Under $H_0$, test statics

$$t = \frac{b_2}{S_{b_2}}$$

$$= \frac{3}{0.8}$$

$$\therefore t_{cal.} = 3.75$$

* Here, level of sig $(\alpha) = 0.05$

$$d.f = n - k - 1$$
$$= 20 - 2 - 1$$
$$= 17$$

From

* t-table

$$t_{tab.} = 2.11$$

* Decision :-

Since $t_{cal.} > t_{tab.}$; we reject $H_0$ & accept $H_1$ with the conclusion that second independent variable is significant.

Q16

→ Here we have

$\Sigma x_1 = 60$

$\Sigma x_2 = 108$

$\Sigma x_3 = 162$

$\Sigma x_1^2 = \cancel{3600}\ 642$

$\Sigma x_2^2 = \cancel{11236}\ 1994$

$\Sigma x_3^2 = 4480$

$\Sigma x_1 x_2 = 1125$

$\Sigma x_1 x_3 =$

Q18) Here we have

$n = 30$, $TSS = \Sigma(y-\bar{y})^2 = 397.6$, $SSE = \Sigma(y-\hat{y})^2 = 23.5$,

$TSS = SSR + SSE$

$397.6 = SSR + 23.5$

$SSR = 374.1$

$k = 2$

Now,

$$MSSR = \frac{SSR}{df}$$

$$= \frac{SSR}{2}$$

$$= \frac{374.1}{2}$$

$$= 187.05$$

$$MSSE = \frac{SSE}{df}$$

$$= \frac{23.5}{30-2-1}$$

$$= 3.357$$

For overall significance

Here,

we set up hypothesis as,

$H_0: B_i = 0$ ie model is insignificant

Vs $H_1: B_i \neq 0$ (TTT) ie model is significant

Under $H_0$, test statistics

$$F = \frac{MSSR}{MSSE} = \frac{187.05}{3.357}$$

$F_{cal} = 55.719$

(*) Here level of significance ($\alpha$) = 5% = 0.05

$df = (k, n-k-1)$
$= (2, 7)$

(*) From $F_{table}$

$F_{tab} = 4.74$

(*) Decision:-
Since $F_{cal} > F_{tab}$ we reject $H_0$ & $H_1$ with the conclusion that regⁿ model is significant

Q.19.

Here we have,

$y = 5 + 18x_1 + 20x_2$ $\quad b_1 = 18$
$\quad b_2 = 20$

$n = 28$, $k = 2$

$TSS = 250$, $SSE = 100$

$SSR = TSS - SSE = 250 - 100 = 150$

$Sb_1 = 3.2$

$Sb_2 = 5.5$

For the test of significance of $X_2$ (2nd Independent Variable)

Here we set up hypothesis as

$H_0: \beta_2 = 0$ ie $X_2$ is insignificant
vs $H_1: \beta_2 \neq 0$ (TIT) ie $X_2$ is significant

Under $H_0$ test static

$$t = \frac{b_2}{Sb_2} = \frac{20}{5.5}$$

$$t_{cal} = 3.64$$

* Here level of significance($\alpha$) = 1% = 0.01

$$d.f = n-k-1$$
$$= 28-2-1$$
$$= 25$$

* From t - table

$$t_{tab} = 2.787$$

* Decision:-

Since $t_{cal} > t_{tab}$ we reject $H_0$ & accept $H_1$ with the conclusion that $X_2$ is significant.

For overall significant of model

Here, we setup hypothesis as,

$H_0: \beta_i = 0$ i.e model is insignificant

vs $H_1: \beta_i \neq 0$ (ITT) ie model is significant

Under $H_0$, test statistic

$$F = \frac{MSSR}{MSSE} = \frac{\frac{SSR}{k}}{\frac{SSE}{n-k-1}} = \frac{\frac{150}{2}}{\frac{100}{25}} = \frac{75}{4}$$

$F_{calc} = 18.75$

Here level of significance ($\alpha$) = 5% = 0.05

$df = (k, n-k-1) = (2, 28-2-1) = (2, 25)$

* From F-table

$F_{tab} = 3.39$

* Decision:-

Since $F_{cal} > F_{table}$, we reject $H_0$ & accept $H_1$ with the conclusion that model is significant.

Q20.

→ Here we have

$\Sigma X_1 = 13$

$\Sigma X_2 = 11$

$\Sigma X_3 = 51$

$\Sigma X_1^2 = 63$

$\Sigma X_2^2 = 95$

$\Sigma X_1 X_3 = 77$

$\Sigma X_2 X_3 = 136$

$\Sigma X_1 X_2 = -240$

$n = 10$

$\Sigma X_3^2 = 450$

(i) Regression equation of $X_3$ on $X_1$ and $X_2$ is

$$X_3 = b_0 + b_1 X_1 + b_2 X_2 \quad \text{— (i)}$$

To estimate $b_0$, $b_1$ & $b_2$

$$\Sigma X_3 = n b_0 + b_1 \Sigma X_1 + b_2 \Sigma X_2$$

$$51 = b_0 10 + b_1 13 + b_2 11 \quad \text{— (ii)}$$

$$\Sigma X_3 X_1 = b_0 \Sigma X_1 + b_1 \Sigma X_1^2 + b_2 \Sigma X_1 X_2$$

$$77 = b_0 13 + b_1 63 + b_2 (-240) \quad \text{— (iii)}$$

$$\Sigma X_3 X_2 = b_0 \Sigma X_2 + b_1 \Sigma X_1 X_2 + b_2 \Sigma X_2^2$$

$$136 = b_0 11 + b_1 (-240) + b_2 95 \quad \text{— (iv)}$$

Using Cramer's Rule

| coeff. of $b_0$ | coeft. of $b_1$ | coeft of $b_2$ | Const. |
|---|---|---|---|
| 10 | 13 | 11 | 51 |
| 13 | 63 | -240 | 77 |
| 11 | -240 | 95 | 136 |

$$D = \begin{vmatrix} 10 & 13 & 11 \\ 13 & 63 & -240 \\ 11 & -240 & 95 \end{vmatrix}$$

$$= 10\left(63 \times 95 - (-240)\times(-240)\right) - 13(13\times 95 - (-240)\times 11) + 11(13 \times (-240) - 63 \times 11)$$

$$= \cancel{-524582}\ -608468$$

$$D_1 = \begin{vmatrix} 51 & 13 & 11 \\ 77 & 63 & -240 \\ 136 & -240 & 95 \end{vmatrix}$$

$$= -3448308$$

$$D_2 = \begin{vmatrix} 10 & 51 & 11 \\ 13 & 77 & -240 \\ 11 & 136 & 95 \end{vmatrix}$$

$$= 212056$$

$$D_3 = \begin{vmatrix} 10 & 13 & \cancel{11}\ 51 \\ 13 & 63 & \cancel{-240}\ 77 \\ 11 & -240 & \cancel{95}\ 136 \end{vmatrix}$$

$$= 64044$$

Now,

$$b_0 = \frac{D_1}{D}$$

$$= \frac{-3449308}{-608468}$$

$$= 5.67$$

$$b_1 = \frac{D_2}{D}$$

$$= \frac{212056}{-608468}$$

$$= -0.35$$

$$b_3 = \frac{D_3}{D}$$

$$= \frac{64044}{-608468}$$

$$= -0.11$$

Subsituiting value in equation I we get

$$X_3 = 5.67 - 0.35 X_1 - 0.11 X_2$$

ii) when $X_1 = 1$ & $X_2 = 4$, predict $X_3$

→ $X_3 = 5.67 - 0.35 \times 1 - 0.11 \times 4$

$= 4.88$

iii) Compute TSS, SSR and SSE

~~TSS~~ Since $TSS = \sum (Y - \bar{Y})^2$ in term of Y, but we need in term of $X_3$ so

$$TSS = \sum (X_3 - \bar{X}_3)^2$$

$$= \sum X_3^2 - n \bar{X}_3^2$$

$$\cancel{= 450 - 10x}$$

Q21. Soln:-

Here, we have

$\sum(y-\bar{y})^2 = 34050$

$\sum(y-\hat{y})^2 = 365.7$

$\sum x_1x_2 = 5779$

$\sum yx_2 = 6796$

$\sum yx_1 = 40830$

$\sum y^2 = 48139$

$\sum x_1^2 = 3483$

$\sum x_2^2 = 976$

$\sum y = 753,$

$\sum x_1 = 643$

$\sum x_2 = 106$

$n = 12$

(i) Regn eqn of y on $x_1$ & $x_2$

$$y = a + b_1x_1 + b_2x_2 \text{ --- (i)}$$

Normal eqn to solve a, $b_1$ & $b_2$ are

$\sum y = na + b_1\sum x_1 + b_2\sum x_2$ — (ii)

$\sum yx_1 = a\sum x_1 + b_1\sum x_1^2 + b_2\sum x_1x_2$ — (iii)

$\sum yx_2 = a\sum x_2 + b_1\sum x_1x_2 + b_2\sum x_2^2$ — (iv)

Now, substituting the value of various sum in Normal eqn

$753 = 12a + 643b_1 + 106b_2$
$40830 = 643a + 3483b_1 + 5779b_2$
$6796 = 106a + 5779b_1 + 976b_2$

By using Cramers rules

| coefft of a | coefft. of $b_1$ | coefft. $b_2$ | const |
|---|---|---|---|
| 12 | 643 | 106 | 753 |
| 643 | 3483 | 5779 | 40830 |
| 106 | 5779 | 976 | 6796 |

$$D = \begin{vmatrix} 12 & 643 & 106 \\ 643 & 3483 & 5779 \\ 106 & 5779 & 976 \end{vmatrix}$$

$= -14860244$

$$D_1 = \begin{vmatrix} 753 & 643 & 106 \\ 40830 & 3483 & 5779 \\ 6496 & 5779 & 976 \end{vmatrix}$$

$= -456105665$

$$D_2 = \begin{vmatrix} 12 & 753 & 106 \\ 623 & 40830 & 5779 \\ 106 & 6796 & 976 \end{vmatrix}$$

$= 57358$

$$D_3 = \begin{vmatrix} 12 & 643 & 753 \\ 648 & 3483 & 40830 \\ 106 & 5779 & 6796 \end{vmatrix}$$

$$= -54277141$$

Now,

$$a = \frac{D_1}{D}$$

$$= \cancel{14860244} \quad \frac{-456105665}{-14860244}$$

$$= 30.69$$

$$b_1 = \frac{D_2}{D}$$

$$= \underline{57358}$$

$$= -0.0038$$

$$b_2 = \frac{-54277141}{-14860244}$$

$$= 3.6525$$

Hence fitted multiple regn eqn of y on $X_1$ & $X_2$ is

$$y = 30.693 - 0.0038X_1 + 3.6525X_2$$

(b) we know,

$$SE = \sqrt{MSSE} \qquad MSSE = \frac{SSE}{n-k-1}$$

$$= \sqrt{\frac{SSE}{n-k-1}}$$

$$= \sqrt{\frac{365.7}{12-2-1}}$$

$$SE = 6.37$$

(c) Here, coefft. of Determination

$$R^2 = \frac{SSR}{TSS} \qquad \text{Here } SSR = TSS - SSE$$

$$= 3450 - 365.7$$

$$SSR = 3084.3$$

$$= \frac{3084.3}{3450}$$

$$= 0.894 \text{ or } 89.4\%$$

which means 89.4% of total variation of dependent variable is due to variation of independent variable $X_1$ & $X_2$

Q.24 Calculation table for various sums

| $x_1$ | $x_2$ | $y$ | $x_1^2$ | $x_2^2$ | $y^2$ | $yx_1$ | $yx_2$ | $x_1x_2$ |
|---|---|---|---|---|---|---|---|---|
| 39 | 8 | 7 | 1521 | 64 | 49 | 2873 | 56 | 312 |
| 52 | 6 | 6 | 2704 | 36 | 36 | 312 | 36 | 312 |
| 49 | 7 | 8 | 2401 | 49 | 64 | 392 | 56 | 343 |
| 46 | 12 | 10 | 2116 | 144 | 100 | 460 | 120 | 552 |
| 61 | 9 | 9 | 3721 | 81 | 81 | 549 | 81 | 549 |
| 35 | 6 | 5 | 1225 | 36 | 25 | 175 | 30 | 210 |
| 25 | 7 | 3 | 625 | 49 | 9 | 75 | 21 | 175 |
| 55 | 4 | 4 | 3025 | 16 | 16 | 220 | 16 | 220 |
| $\Sigma x_1 =$ 362 | $\Sigma x_2 =$ 59 | $\Sigma y =$ 52 | $\Sigma x_1^2 =$ 17338 | $\Sigma x_2^2 =$ 475 | $\Sigma y^2 =$ 380 | $\Sigma yx_1 =$ 2456 | $\Sigma yx_2 =$ 416 | $\Sigma x_1x_2 =$ 2673 |

$\Sigma x_1 = 362$ $\quad n = 8$

$\Sigma x_2 = 59$

$\Sigma y = 52$

$\Sigma x_1^2 = 17338$

$\Sigma x_2^2 = 475$

$\Sigma y^2 = 380$

$\Sigma yx_1 = 2456$

$\Sigma yx_2 = 416$

$\Sigma x_1x_2 = 2673$

(e)

$$y = a + b_1 x_1 + b_2 x_2 \quad \leftarrow (e)$$

$\Sigma y = na + b_1 \Sigma x_1 + b_2 \Sigma x_2$ --- (i)
$\Sigma y x_1 = a \Sigma x_1 + b_1 \Sigma x_1^2 + b_2 \Sigma x_1 x_2$ --- (ii)
$\Sigma y x_2 = a \Sigma x_2 + b_1 \Sigma x_1 x_2 + b_2 \Sigma x_2^2$ --- (iii)

Now,
Subsituting the values in eqn

$52 = 8a + b_1 362 + b_2 59$
$2456 = 362a + 17338 b_1 + b_2 2673$
$416 = 59a + 2673 b_1 + 475 b_2$

Now
using cramers rule

| coeff of a | coeff of $b_1$ | coeff of $b_2$ | const |
|---|---|---|---|
| 8 | 362 | 59 | 52 |
| 362 | 17338 | 2673 | 2456 |
| 59 | 2673 | 475 | 416 |

$$D = \begin{vmatrix} 8 & 362 & 59 \\ 362 & 17338 & 2673 \\ 59 & 2673 & 475 \end{vmatrix}$$

$= \cancel{-10757842}$ 305358

$$D_1 = \begin{vmatrix} 52 & 362 & 59 \\ 2456 & 17338 & 2673 \\ 416 & 2673 & 475 \end{vmatrix}$$

$= \cancel{-1308239772}$
$-1279972$

$$D_2 = \begin{vmatrix} 8 & 52 & 59 \\ 362 & 2456 & 2973 \\ 59 & 416 & -475 \end{vmatrix}$$

$$= 32012$$

$$D_3 = \begin{vmatrix} 8 & 362 & 52 \\ 362 & 17338 & 2456 \\ 59 & 2673 & 416 \end{vmatrix}$$

$$= 246272$$

$$a = \frac{D_1}{D}$$

$$= \cancel{305}\ \cancel{\frac{-130823872}{305358}} \qquad \frac{-1279992}{305358}$$

$$= \cancel{-428.48} \qquad = -4.19$$

$$b_1 = \frac{D_2}{D}$$

$$= \frac{32012}{305358}$$

$$= 0.1048$$

$$b_2 = \frac{D_3}{D}$$

$$= \frac{246272}{305358}$$

$$= 0.8065$$

Placing all value in eqn (i)

$$y = -4.19 + 0.10 x_1 + 0.80 X_2$$

(ii) Standard error of estimate.

$$s = \sqrt{\frac{\Sigma y^2 - a\Sigma y - b_1 \Sigma y x_1 - b_2 \Sigma y x_2}{n-k-1}}$$

$$= \sqrt{\frac{380 - (-4.19) \times 52 - 0.10 \times 2456 - 0.80 \times 416}{5}}$$

$$= \cancel{0.59}\ 0.99$$

(iii)

to find the weight gain of pig (y) when initial age $(x_2) = 9$ & initain weight $(x_1)$ is 48.

Now,

we know,

$$y = a + b_1 x_1 + b_2 x_2$$

$$= -4.19 - 0.1048 \times 48 + 0.8062 \times 9$$

$$= 8.09 \text{ pound}$$

2.

| $x_1$ | $x_2$ | $x_3$ | $x_1^2$ | $x_2^2$ | $x_3^2$ | $x_1x_2$ | $x_1x_3$ |
|---|---|---|---|---|---|---|---|
| 5 | 12 | 51 | 25 | 144 | 2601 | 60 | 255 |
| 7 | 20 | 55 | 49 | 400 | 3025 | 140 | 385 |
| 8 | 30 | 58 | 64 | 900 | 3364 | 240 | 464 |
| 6 | 40 | 60 | 36 | 1600 | 3600 | 240 | 360 |
| 10 | 33 | 70 | 100 | 1089 | 4900 | 330 | 700 |
| 9 | 25 | 66 | 81 | 625 | 4356 | 225 | 594 |
| $\Sigma x_1 = 45$ | $\Sigma x_2 = 160$ | $\Sigma x_3 = 360$ | $\Sigma x_1^2 = 355$ | $\Sigma x_2^2 = 4758$ | $\Sigma x_3^2 = 21848$ | $\Sigma x_1x_2 = 1235$ | $\Sigma x_1x_3 = 2758$ |

$\Sigma x_1 = 45$

$\Sigma x_2 = 160$

$\Sigma x_3 = 360$

$\Sigma x_1^2 = 355$

$\Sigma x_2^2 = 4758$

$\Sigma x_3^2 = 21848$

$\Sigma x_1 x_2 = 1235$

$\Sigma x_1 x_3 = 2758$

$\Sigma x_2 x_3 = 9812$

Now,

Regression Equation of $x_1$ on $x_2$ & $x_3$

$$x_1 = a + b_1 x_2 + b_2 x_3$$

Chapter-1

## * Sampling Distribution

→ It is the distribution followed by a certain statistics computed from sample

## * Standard Error

→ It is the standard ~~distribution~~ deviation of a statistics of sampling distribution

Eg:-

* $SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$ (For infinite N or SRSWR)

$= \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}$ (for finite N or SRSWOR)

* $SE(P) = \sqrt{\frac{PQ}{n}}$ (For Infinite N or SRSWR)

$= \sqrt{\frac{PQ}{n}} \times \sqrt{\frac{N-n}{N-1}}$ (For finite N or SRSWOR)

## * ESTIMATION

It is the process of predicting the value of population parameter by the study of sample statistics.

There are two types of estimation

(i) Point estimation.
(ii) Interval estimation
- Confidence Interval
- Fiducial Limits

$(1-\alpha)\%$ Confidence Interval

* For pop$^n$ mean (Large Sample)

$(1-\alpha)\%$ C.I. for $\mu = \bar{x} \pm z_\alpha \times S.E.(\bar{x})$

* For pop$^n$ proportion

$(1-\alpha)\%$ C.I. for $P = p \pm z_\alpha \times SE(p)$

Sample size determination

A) For the estimation of pop$^n$ mean

$$n = \left( \frac{z_\alpha \times \sigma}{E} \right)^2$$

where, $\alpha$ = Risk or level of significance
$1-\alpha$ = confidence level or Probability
$z_\alpha$ = Table value of $z$ at $\alpha\%$ level of sig.
$E = |\bar{x} - \mu|$ = Max. permissable Error
$\sigma$ = pop$^n$ S.D

B) For the estimation of pop$^n$ proportion

$$n = \left(\frac{z_\alpha}{E}\right)^2 \times PQ$$

where, $E = |p - P|$ = Max permissable Error

$P$ = popn prop

$Q = 1 - P$

Note:-

① If '$\alpha$' is not given, we use $\alpha = 5\%$

② For almost sure, certainly i.e, probable, i.e we should take $z_\alpha = 3$.

③ If value of $P$ is not given, then we should take

$$P = Q = 1/2$$

Q5

Soln:-

Here we have

Sample size $(n) = 25$

Population size $(N) = 150$

Population SD $(\sigma) = 10$

$SE(\bar{x}) = ?$

Now,

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}$$

$$= \frac{10}{\sqrt{25}} \times \sqrt{\frac{150-25}{150-1}}$$

$S.E.(\bar{x}) = 1.83$

Q.8

→ Soln:-

Here we have,

Sample size (n) = 25

Population size (N) = 300

No. of not correctly sent message (x) = 15

$\Rightarrow$ Sample proportion(p) $= \frac{x}{n} = \frac{15}{25} = 0.6$

$\Rightarrow q = 1 - p = 1 - 0.6 = 0.4$

Now,

$$S.E(p) = \sqrt{\frac{PQ}{n}} \times \sqrt{\frac{N-n}{N-1}}$$

$$= \sqrt{\frac{0.6 \times 0.4}{25}} \times \sqrt{\frac{300-25}{300-1}}$$

$\therefore S.E.(P) = 0.094$

Q.7)

→ Soln:-

Here we have :

Sample size (n) = 20

Population size (N) = 75

No. of defecte tive units (x) = 12.

We know,

$\Rightarrow$ Sample proportion(p) $= \frac{x}{n} = \frac{12}{20} = 0.6$

$\Rightarrow q = 1 - p = 1 - 0.6$

$= 0.4$

Now,
$$SE(P) = \sqrt{\frac{PQ}{n}} \times \sqrt{\frac{N-n}{N-1}}$$

$$= \sqrt{\frac{0.6 \times 0.4}{20}} \times \sqrt{\frac{75-20}{75-1}}$$

$$= 0.094$$

Q9

Here we have

Town - A

$n_1 = 70,\ x_1 = 42$

$$P_1 = \frac{x_1}{n_1}$$

$$= \frac{42}{70}$$

$$= 0.6$$

Town - B

$n_2 = 100$ & $x_2 = 59$

$$P_2 = \frac{x_2}{n_2}$$

$$= \frac{59}{100}$$

$$= 0.59$$

Now,

combined proportion $(P) = \frac{x_1 + x_2}{n_1 + n_2}$

$$= \frac{42 + 59}{100 + 70}$$

$$= 0.594$$

$$\therefore q = 1 - P = 1 - 0.594$$

$$= 0.4059$$

Now,

$$SE(P_1 - P_2) = \sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$= \sqrt{0.594 \times 0.4059\left(\frac{1}{70} + \frac{1}{100}\right)}$$

$$= 0.0765$$

Q10

→ Here we have,

For 1st

$n_1 = 500$

$\bar{x}_1 = 0.8$

$(s_1)\sigma_1 = 0.1$

For 2nd

$n_2 = 300$

$\bar{x}_2 = 0.5$

$(s_2)\sigma_2 = 0.08$

Now

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$= \sqrt{\frac{(0.1)^2}{500} + \frac{(0.08)^2}{300}}$$

$$= 0.0064$$

Q 11

→ Here we have

Sample size $(n) = 64$

Pop$^n$ standard deviation $(\sigma) = 20$

mean $(\bar{x}) = 80$

C.I. $(1-\alpha) = 95\%$

Here

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

$$= \frac{20}{\sqrt{64}}$$

$$= 2.5$$

Again,

$$p(0 < z < z_\alpha) = \frac{1-\alpha}{2}$$

$$= \frac{0.95}{2}$$

$$= 0.475$$

From SNT, $z_\alpha = 1.96$

Now,

95% CI for pop$^n$ mean$(\mu) = \bar{x} \pm z_\alpha \times SE(\bar{x})$

$= 80 \pm 2.5 \times 1.96$

$= 80 \pm 4.9$

∴ 95% CI for pop$^n$ mean $(\mu) = (75.1 - 84.9)$

2nd part

Here,

sample size $(n) = 256$

Then,

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{256}} = 1.25$$

Now, 95% CI for $pop^n$ mean $(\mu) = \bar{x} \pm z_\alpha \times SE(\bar{x})$

$= 80 \pm 1.25 \times 1.96$

$= 80 \pm 2.45$

95% CI for $pop^n$ mean $(\mu) = (77.55 - 82.45)$

Q.13

→ Here we have

Sample size $(n) = 70$

No. of success $(x) = 28$

Then,

$$\text{Sample prop}(P) = \frac{x}{n} = \frac{28}{70} = 0.4$$

$$q = 1 - p = 1 - 0.4 = 0.6$$

Now,

$$SE(P) = \sqrt{\frac{PQ}{n}}$$

$$= \sqrt{\frac{0.4\times 0.6}{70}}$$

$$= 0.0586$$

Here,

$CI\,(1-\alpha) = 90\%$

$= 0.9$

We know

$$P(0<z<z_\alpha) = \frac{1-\alpha}{2}$$

$$= \frac{0.9}{2}$$

$$= 0.45$$

From SNT,

$z_\alpha = 1.645$

Now,

90% CI for pop$^n$ prop(P) $= P \pm z_\alpha \times S.E(P)$

$= 0.4 \pm 1.645 \times 0.0586$

$= 0.4 \pm 0.0964$

$\therefore$ 90% CI for pop$^n$ prop(P) $= (0.3036 - 0.4964$

Q14 Here we have,

Sample size $(n) = 400$

No. of defective $(x) = 20$

Sample prop $(p) = \frac{x}{n}$

$= \frac{20}{400}$

$= 0.05$

$\therefore q = 1 - p$

$= 1 - 0.5$

$= \cancel{0.5}\ 0.95$

Now,

$SE(p) = \sqrt{\frac{pq}{n}}$

$= \sqrt{\frac{0.05 \times 0.95}{400}}$

$= \cancel{0.025}\ 0.0109$

Here,

$CI\ (1-\alpha) = 99\%$

$= 0.99$

We know,

$P(0 < z < z_\alpha) = \frac{1-\alpha}{2}$

$= \frac{0.99}{2}$

$= 0.495$

From SNT,

$z_\alpha = 2.575$

99 % CI for popⁿ prop $= p \pm z_\alpha \times SE(p)$

$\cancel{= 0.5 \pm 2.575 \times 0.02}$

$\cancel{= (0.5 \pm 0.0644)}$

$= 0.05 \pm 2.575 \times 0.0085$

$= 0.05 \pm 0.0281$

$= 0.05 \pm 0.0281$

$= \cancel{0.0781}$

$= (0.0219 - 0.078)$

99 % CI for popⁿ prop $= 2.2\% - 7.8\%$

Q15

→ Soln:-

Here we have

Sample size $(n) = 400$

No. of tested good $(x) = 80$

Sample prop $p\left(\frac{x}{n}\right) = \frac{80}{400}$

$p = 0.2$

$q = 1 - p$

$= 1 - 0.2$

$= 0.8$

Now,

$$SE(p) = \sqrt{\frac{pq}{n}}$$

$$= \sqrt{\frac{0.2 \times 0.8}{400}}$$

$= 0.02$

$CI\ (1-\alpha) = 95\%$

$= 0.95$

We know,

$P(0 < Z < z_\alpha) = \frac{1-\alpha}{2}$

$= 0.475$

From SNT,

$z_\alpha = 1.96$

95% CI for popⁿ prop $= P \pm z_\alpha \times SE(P)$

$= 0.2 \pm 1.96 \times 0.02$

$= 0.2 \pm 0.0392$

$= (0.1608 - 0.2392$

95% CI for popⁿ prop $= (16\% - 23\%)$

Q20) Soln:-

Here we have,

popⁿ SD $(\sigma) = 3$

Sample size $(n) = ?$

Max. permissable Error $(E) = 0.5$

level of significance $(\alpha) = 5\%$ let,

We know

$P(0 < z < z_\alpha) = \frac{1-\alpha}{2}$

$= \frac{1-0.05}{2}$

$= 0.4750$

From SNT, $z_\alpha = 1.96$

Now,

req'd sample size $(n) = \left(\frac{z_\alpha \times \sigma}{E}\right)^2$

$= \left(\frac{1.96 \times 3}{0.5}\right)^2$

$= 138.29$

$\approx 138$

Q21)

Here we have,

Sample size $(n) = ?$

Max. permissiable Error $(E) = 10\%$

$= 0.1$

For

almost certainty,

$z_\alpha = 3$

prop. $(P) = 0.3$

$Q = 1 - P$

$= 1 - 0.3$

$= 0.7$

We know,

reqd sample size

$$n = \left(\frac{z_\alpha}{E}\right)^2 \times PQ$$

$$= \left(\frac{3}{0.1}\right)^2 \times 0.3 \times 0.7$$

$$n = 189$$

Q22

→ Here we have,

pop$^n$ SD($\sigma$) = 10

sample size (n) = ?

CI $(1-\alpha)$ = 95%

= 0.95

we know,

$$p(0 < z < z_\alpha) = \frac{1-\alpha}{2}$$

$$= \frac{0.95}{2}$$

$$= 0.475$$

From SNT,

$$z_\alpha = 1.96$$

Max. permissable error ($E$) = 2

We know,

$$n = \left(\frac{z_\alpha \times \sigma}{E}\right)^2 = \left(\frac{1.96 \times 10}{2}\right)^2$$

$$= 96$$

ch-2

Z- test (large sample Test (n>30))

* Case-I: Test of significance of sample popⁿ mean

Test - statistic

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

If ($\sigma$) is not given we use sample S.D. (S).

* Case - II :- Test of significant difference in two popⁿ means

Test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

If $S_1^2$ & $S_2^2$ are given then

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

* Case (III) Test of Significance of single ~~propn~~ proportion

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

where, $\hat{p} = \frac{x}{n}$ = Sample prop.

$P$ = ~~Pop~~ popn prop.

$Q = 1 - P$

* Case (IV) Test of significance difference in two popn proportions

Test Statistics

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where, $\hat{p}_1 = \frac{x_1}{n_1}$ (1st sample Proportion)

$\hat{p}_2 = \frac{x_2}{n_2}$ (2nd sample Proportion)

$P$ = Combined proportion

$$= \frac{x_1 + x_2}{n_1 + n_2}$$

$$P = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

& $Q = 1 - P$

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t-test (small sample test $n \leq 30$)

* Case-(I) Test of significance of single pop<sup></sup> mean

Test-statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

if data are given

Here, $s^2 = \frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]$

or,

$$s^2 = \frac{1}{n-1}\left[\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right]$$

If 'S' is given

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n-1}}$$

Here $df = n-1$

* Case-(II) Test of significant difference between two popn means

Test-statistics

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where,

$$S_p^2 = \frac{1}{n_1 + n_2 - 2}\left[n_1 s_1^2 + n_2 s_2^2\right]$$

or, $$S_p^2 = \frac{1}{n_1 + n_2 - 2}\left[\sum x_1^2 - \frac{(\sum x_1)^2}{n_1} + \sum x_2^2 - \frac{(\sum x_2)^2}{n_2}\right]$$

$$= \frac{1}{n_1 + n_2 - 2}\left[\sum d_1^2 - \frac{(\sum d_1)^2}{n_1} + \sum d_2^2 - \frac{(\sum d_2)^2}{n_2}\right]$$

$$= \frac{1}{n_1 - n_2 - 2}\left(\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2\right]$$

$d_1 = x_1 - A$, $d_2 = x_2 - B$

Here $d.f = n_1 + n_2 - 2$

* Case (III) paired t-Test

$$t = \frac{\bar{d}}{sd/\sqrt{n}}$$ If data are given

Here. $$S_d^2 = \frac{1}{n-1}\left[\sum d^2 - \frac{(\sum d)^2}{n}\right]$$

If $sd$ is given then

$$t = \frac{\bar{d}}{sd/\sqrt{n-1}}$$

Here $d = x - y$

$x$ = score before

$y$ = ,, after

Here $df = n-1$

* Case (IV) Test of significance of single correlⁿ coefft

Test - statistics

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Here, $df = n-2$

$(1-\alpha)\%$ CI for popⁿ correlⁿ coefft ($\rho$)

$= r \pm t_{\alpha, n-2} \cdot SE(r)$

Here, $SE(r) = \frac{1-r^2}{\sqrt{n}}$

12 - Mangsir

14.) Soln:-

Here we have,

Sample size $(n) = 400$

Sample mean $(\bar{x}) = 99$

Popⁿ mean $(\mu) = 100$

level of significant $(\alpha) = 5\%$

$= 0.05$

Population SD $(\sigma) = 8$

Here,

we set up hypothesis as

$H_0: \mu = 100$ ie Sample is taken from popⁿ with mean 100.

N/S $H_1: \mu \neq 100$ (TTT) ie sample is taken from popⁿ with mean other than 100.

Under $H_0$, test statistics

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{99 - 100}{8/\sqrt{400}}$$

$$= -2.5$$

$\therefore |z_{cal}| = 2.5$

Here,

level of significance $(\alpha) = 5\%$

$= 0.05$

From table

$z\alpha = 1.96$

$P(0 < z < z\alpha) = \frac{1-\alpha}{2}$

$= 0.475$

<u>Decision:</u> Since $z_{cal} > z_{tab}$; we reject $H_0$ & accept $H_1$ with the conclusion that sample is taken from popⁿ with mean other than 100

16 Soln:

Here we have,

Popn mean ($\mu$) = 1800

" SD ($\sigma$) = 100

sample size (n) = 50

" mean ($\bar{x}$) = 1850

Here, we set up hypothesis as

$H_0$: $\mu = 1800$ i.e claim is invalid

vs $H_1$: $\mu > 1800$ (OTT) ie claim is valid

Under $H_0$, test statistics

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{1850 - 1800}{\frac{100}{\sqrt{\cancel{1800}\ 50}}}$$

$z_{cal.} = 3.53$

Here

level of significance ($\alpha$) = 1%

From table = 0.01

$z_{tab} = 2.33$

| OTT (One-Tailed Test)
| $P(0 < z < z_\alpha) = \frac{1}{2}$ -
| = 0.5-
| = 0.49

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Decision:-

Since $z_{cal} > z_{tab}$; we reject $H_0$ & accept $H_1$ with the conclusion that claim is valid.

17.)

Soln:-

Here we have,

Sample size $(n) = 400$

sample mean $(\bar{x}) = 171.38$

$pop^n$ mean $(\mu) = 171.17$

$pop^n$ S.D $(\sigma) = 3.3$

Here,

we set up hypothesis as

$H_0: \mu = 171.17$ ie sample is taken from given $pop^n$

V/S $H_1: \mu \neq 171.17$ (TT) ie sample is taken from other $pop^n$

under $H_0$, test statistic

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{171.38 - 171.17}{\frac{3.3}{\sqrt{400}}}$$

$$z_{cal} = 1.27$$

Here, level of significance $(\alpha) = 5\%$
$= 0.05$ (...

From table for (TTT)
$z_{tab} = 1.96$

For confidence limit approach

$$CI = \mu \pm z_\alpha \, SE(\bar{x})$$
$$= \mu \pm z_\alpha \times \frac{\sigma}{\sqrt{n}}$$
$$= 171.17 \pm 1.96 \times \frac{3.3}{\sqrt{400}}$$
$$= 171.17 \pm 0.3234$$

$\therefore$ CI $= (170.8466 - 171.4934)$

$\therefore$ Since $\bar{x} \in CL$, So, we accept $H_0$ & reject $H_1$ with the conclusion the sample is taken from given popⁿ:-

21. Soln:-

Here we have,

For first process

$n_1 = 250$, $\bar{x}_1 = 120$

$s_1 = 12$

for second process

$n_2 = 400$, $\bar{x}_2 = 124$

$S_2 = 14$

Here we set up Hypothesis as

$H_0: \mu_1 = \mu_2$ i.e There is no significant difference in two population.

v/s $H_1: \mu_1 \neq \mu_2$ (TTT) ie there is significant difference in two popn.

Under $H_0$, test statistics

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{120 - 124}{\sqrt{\frac{12^2}{250} + \frac{14^2}{400}}}$$

$\therefore |z_{cal}| = |-3.87|$

$= 3.87$

Here,

Level of significance ($\alpha$) = 10%

= 0.01

For two tailed test

p-value = $2 \times P(Z > |z_{cal.}|)$

$= 2 \times P(Z > 3.87)$

$= 2 \times [P(0 < Z < \infty) - P(0 < Z < 3.87)]$

$= 2(0.5 - 0.49995)$

$\therefore$ p-value $= 0.0001$

Decision:- Since p-value $< \alpha$, we reject $H_0$ & accept $H_1$ with the conclusion that there is significant difference in two population.

24) Soln:-

Here, we know,

For first sample — For second sample

$n_1 = 35$, $\bar{x}_1 = 81$ — $n_2 = 36$, $\bar{x}_2 = 76$

$s_1 = 5.2$ — $s_2 = 3.4$

Here we set a hypothesis as,

$H_0: \mu_1 = \mu_2$ ie there is no significant difference in two population

V/S $H_1: \mu_1 \neq \mu_2$ (TTT) ie There is significant difference in two population.

Under Ho, test statistics

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{81 - 76}{\sqrt{\frac{5.2^2}{35} + \frac{3.4^2}{36}}}$$

$\therefore |Z_{cal}| = 4.78$

Here,

level of significant $(\alpha) = 5\%$

$= 0.05$

For two tailed test,

From SNT, $Z_\alpha = 1.96$

Decision:-

Since, $Z_{cal} > Z_{tab}$, we reject $H_0$ & ~~$H_1$~~ accept $H_1$ with the conclusion that there is significant difference in two hypothesis.

25. Soln:-

Here, we have

For First Sample

$n_1 = 80$, $\overline{x_1} = 8.5$

$s_1 = 1.8$

For Second Sample

$n_2 = 80$, $\overline{x_2} = 7.9$

$s_2 = 2.1$

Here, we set up Hypothesis as

$H_0: \mu_1 = \mu_2$ i.e. There is no significant difference in two Drugs

v/s $H_1: \mu_1 \neq \mu_2$ (TTT) i.e. ~~there~~ Secondary drug provides significantly shorter period of relief.

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Under $H_0$, test statistics

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{8.5 - 7.9}{\sqrt{\frac{1.8^2}{90} + \frac{2.1^2}{80}}}$$

$z_{cal} = 1.99$

Here level of sig. $(\alpha) = 5\% = 0.05$
For one tailed test

From SNT $Z_\alpha = 1.645$

For OTT

$$\frac{1-\alpha}{2} = \frac{0.5 - 0.05}{2}$$

$= 0.45$

Decision:

Since $Z_{cal} > Z_{tab}$. We reject $H_0$ and accept $H_1$ with the conclusion that second drug provide significantly shorter ~~provide~~ period of relief.

28.

→ Soln:-

Here we have,

sample size $(n) = 9000$

No. of times dice show 3 or 4 $(x) = 3220$

Sample prop $(p) = \frac{x}{n}$

$= \frac{3220}{9000}$

$= 0.357$

Popn prop $(P) = \frac{2}{6}$

$= \frac{1}{3}$

$= 0.33$

$Q = 1 - P$

$= 1 - 0.33$

$= 0.67$

Here, we set up hypothesis as,

$H_0$: $P = \frac{1}{3}$ ie Dice is unbiased,

v/s $H_1$: $P \neq \frac{1}{3}$ (TTT) ie Dice is biased

Under $H_0$, test statistics

$$z = \frac{p - P}{\sqrt{PQ/n}} = \frac{0.357 - 0.33}{\sqrt{\frac{0.33 \times 0.67}{9000}}}$$

$$z_{cal} = 4.82$$

Here

level of significance $(\alpha) = 5\%$
$= 0.05$

For two tailed test

$Z_{tab} = 1.96$ (From table)

Decision:-

Since $Z_{cal} > Z_{tab}$, we reject $H_0$ & accept $H_1$ with the conclusion that dice is biased.

30.)

Soln:-

Here we have,

Sample size $(n) = 250$

No. of Student who submitted assignment on time $(x) = 250 - 15$
$= 235$

Sample prop $(p) = \frac{x}{n} = \frac{235}{250}$

Popn prop $(P) = 98\%$
$= 0.98$

$Q = 1 - 0.98$
$= 0.02$

Here, we set up Hypothesis as

$H_0: P \geq 98\%$ ie Claim is valid

V/S $H_1: P < 98\%$ (OTT) ie Claim is invalid

Under Ho, Test static

$$z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$$

$$= \frac{0.94-0.98}{\sqrt{\frac{0.98 \times 0.02}{250}}}$$

$$= -4.51$$

$\therefore |z_{cal.}| = 4.51,$

Here

level of significance $(\alpha) = 10\%$
$= 0.1$

For two tailed test

$z_{tab} = 1.28$ (From table)

Decision:

Since $z_{cal} > z_{tab}$, we reject $H_0$ accept $H_1$ with the conclusion claim is invalid.

34.

Soln:

Here we have,

Before overhauling

$n_1 = 400$, $x_1 = 20$

$p_1 = \frac{x_1}{n_1} = \frac{1}{20}$

After overhauling

$n_2 = 300$, $x_2 = 10$

$p_2 = \frac{x_2}{n_2} = \frac{1}{30}$

$$P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{20+10}{400+300} = \frac{3}{70}$$

$$Q = 1 - P = \frac{67}{70}$$

Here,

we set up hypothesis as,

$H_0: P_1 = P_2$ ie Machine has not been improved

v/s $H_1: P_1 > P_2$ (OTT) ie Machine has been improved.

Under $H_0$, test statistics,

$$z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$z_{cal.} = 1.1$

Here,

level of significance $(\alpha) = 5\% = 0.05$

For OTT

$Z_{tab} = 1.645$

Decision:

Since $Z_{cal} < Z_{tab}$, we accept $H_0$ and ~~ge~~ reject $H_1$ with the conclusion that Machine has not been improved.

31)

→ Soln:-

Here we have

popn prop $(P) = 40\% = 0.4$   $Q = 1 - P = 0.6$

Sample size $(n) = 150$

No. of threatening calls $(x) = 49$

Level of significance $(\alpha) = 1\%$

Sample prop $(p) = \frac{x}{n}$

$= \frac{49}{150}$

$\cancel{= 0.326}$

Here, we set hypothesis

$H_0$: $P \geq 40\%$ claim is valid

V/S $H_1$: $P < 40\%$ (OTT) ie claim is invalid

Under $H_0$ test statistics

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{\frac{49}{150} - 0.4}{\sqrt{\frac{0.4 \times 0.6}{150}}} = -1.83

$|z| = 1.83$

P-value $= P(z > |z_{cal}|)$

$= P(z > 1.83)$

$= P(0 < z < \infty) - P(0 < z < 1.83)$

$= 0.5 - 0.4664$

$= 0.034$

Here

Level of significance $(\alpha) = 1\%$

$= 0.01$

Decision

P-value $= 0.034 > \alpha = 0.01$ so we accept $H_0$ & reject $H_1$ with the conclusion that claim is valid.

32.

→ soln:-

Here we have,

Sample size $(n) = 600$

No. of people who liked

Consume tea $(x) = 310$

Sample porpn $(p) = \frac{x}{n}$

$= \frac{310}{600}$

$= 0.517$

$Q = 1 - p$

$= 1 - 0.517$

Q = 0.485

38.

Here we have

Sample size $(n) = 12$

Sample mean $(\bar{x}) = 50$

Sample S.D $(s) = 30$

Pop$^n$ mean $(\mu) = 51.2$

Here we set up hypothesis as

$H_0: \mu = 51.2$ i.e Avg. rainfall is 51.2 mm

U/S $H_1: \mu < 51.2$ (OTT) ie Avg rainfall is less than 51.2 mm

Under $H_0$ test statistics is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$$= \frac{50 - 51.2}{30/\sqrt{12-1}}$$

$|t_{cal}| = |-0.133|$

$|t_{cal.}| = 0.133$

Here, level of sig$(\alpha) = 10\% = 0.1$

d.f $= n-1$

$= 12-1$

$= 11$

For one tailed test,

~~$t_{tab}$~~ $t_{tab} = 1.363$

Decision:-

Since $t_{cal} < t_{tab}$ : we accept $H_0$ & reject $H_1$ with the conclusion that Avg. rainfall is 51.2 mm.

44) Calculation for $\bar{X}$ & S

| X | d' | $d'^2$ |
|---|---|---|
| 35 | -1 | 1 |
| 20 | -4 | 16 |
| 30 | -2 | 4 |
| 45 | 1 | 1 |
| 60 | 4 | 16 |
| 40 | 0 | 0 |
| 25 | -3 | 9 |
| 50 | 2 | 4 |
| | $\Sigma d' = -3$ | $\Sigma d'^2 = 51$ |

Here, $d' = \frac{X-A}{h}$ , $n=8$

$A = 40$ , $h = 5$

| | | |
|---|---|---|
| 65 | 5 | 25 |
| 40 | 0 | 0 |
| 25 | -3 | 9 |
| 50 | 2 | 4 |
| | $\Sigma d' = 2$ | $\Sigma d'^2 = 76$ |

$n = 10$

Here, $d' = \frac{x - A}{h}$

$A = 40$, $h = 5$

For $\bar{X}$

$$\bar{X} = A + \frac{\Sigma d'}{n} \times h$$

$$= 40 + \left(\frac{2}{10}\right) \times 5$$

$$= 41$$

Chi 13k kruskal Kolmogorov

$s^2$

$$s^2 = h^2 \times \left\{ \frac{1}{n-1} \left( \Sigma d'^2 - \frac{(\Sigma d')^2}{n} \right) \right\}$$

$$= 5^2 \times \left\{ \frac{1}{10-1} \left( 76 - \frac{2^2}{10} \right) \right\}$$

$s^2 = 210$

$s = 14.49$

popn mean $(\mu) = 30$

Sample mean $(\bar{X}) = 41$

Sample SD $(s) = 14.49$

Sample size $(n) = 10$

Here we set up hypthesis as

$H_0 : \mu = 30$ ie popn average is 30 mins

V/S $H_1 : \mu > 30$ (OTT) ie popn " is more than 30m

Under $H_0$, test statistics is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$= \frac{41-30}{14.49/\sqrt{10}}$$

$t_{cal} = 2.4$

Here, level of significance$(\alpha) = 5\% = 0.05$

$df = n-1$

$= 10-1$

$= 9$

For one tailed test,

$t_{tab} = 1.833$

Decision:

Since $t_{cal} > t_{tab}$; we reject $H_0$ & accept $H_1$ with the conclusion that $pop^n$ average is more than 30 mins.

18) Calculation table for $\bar{X}_1$, $\bar{X}_2$ & $Sp^2$

| $x_1$ | $d_1$ | $d_1^2$ | $x_2$ | $d_2$ | $d_2^2$ |
|---|---|---|---|---|---|
| 18 | -18 | 324 | 29 | -1 | 1 |
| 20 | -16 | 256 | 28 | -2 | 4 |
| 36 | 0 | 0 | 26 | -4 | 16 |
| 50 | 14 | 196 | 35 | 5 | 25 |
| 49 | 13 | 160 | 30 | 0 | 0 |
| 36 | 0 | 0 | 44 | 14 | 196 |
| 34 | -2 | 4 | 46 | 16 | 256 |
| 49 | 13 | 169 | | | |
| 41 | 5 | 25 | | | |

$\Sigma d_1 = 9$

$\Sigma d_1^2 = 1143$

$\Sigma d_2 = 28$

$\Sigma d_2^2 = 498$

Here, $n_1 = 9$, $n_2 = 7$

$d_1 = x_1 - A$, $d_2 = x_2 - B$

let $A = 36$, $B = 30$

Here,

$$\overline{x_1} = A + \frac{\Sigma d_1}{n_1} = 37$$

$$\overline{x_2} = B + \frac{\Sigma d_2}{n_2} = 34$$

& $$S_p^2 = \frac{1}{n_1 + n_2 - 2}\left[\Sigma d^2 - \frac{(\Sigma d_1)^2}{n_1} + \Sigma d_2^2 - \frac{(\Sigma d_2)^2}{n_2}\right]$$

$$= 108.57$$

$S_p = 10.42$

Here

level of significance $(\alpha) = 5\%$

$= 0.05$

Ch-3

* Run Test

It is the non-parametric test used to test the randomness or uniformity of data.

* RUN:-

It is a sequence of similar symbol

For Small sample ($n_1, n_2 \leq 20$)

Test statistics

No. of runs ($r$)

Critical value

We should take values of $\underline{r}$ & $\overline{r}$ from run-table on the basis of $n_1$ & $n_2$

where,

$\underline{r}$ = lower bound of $r$

$\overline{r}$ = upper " " "

$n_1$ = No. of 1st symbol

$n_2$ = No. of 2nd symbol

Decision:- If $r \in (\underline{r}, \overline{r})$, we accept $H_0$ & reject $H_1$

For large sample ($n_1$ or $n_2 > 20$)

Test statistics

$$z = \frac{r - \mu_r}{\sigma_r}$$

where, $\mu_r = \dfrac{2n_1 n_2}{n_1 + n_2} + 1$

$$\sigma_r^2 = \frac{2n_1 n_2 (2n_1 n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}$$

Note

① If numerical data is given then we compute $M_d$ of given data & assign symbol as,

$A =$ If $x_i < M_d$

$B =$ If $x_i > M_d$

Omit the data if $x_i = M_d$

② For $\alpha \neq 5\%$, we use $Z$ - test

* Question solving

9)

→ Soln:-

Here we have given set of data

HTTHTHHHT HHTT HTHTHH THTTH THHHTHTH

Here

$n_1 = 17$ ← (no. of H)

$n_2 = 14$ ← (no of T)

no. of runs $(r) = 23$

Here, we setup hypothesis as

$H_0$: Sequence is in random order

vs $H_1$: Sequence is not in random order

Under $H_0$, test statistics,

No. of runs $(r) = 23$

Table of critical value
for level of significance $(\alpha) = 5\% = 0.05$
$n_1 = 17$ , $n_2 = 14$
$\underline{r} = 10$
$\overline{r} = 23$

Decision:-
If $r \notin (\underline{r}, \overline{r})$, we reject $H_0$ & accept $H_1$, with the conclusion the sequence is not in random order.

13)
→Soln:-
Here, we have
65.3, 67.7, 68.4, 71, 70.2, 67, 69.8, 64, 67.6, 71.5
64.4, 63.6, 66.8, 70.4, 69
Here we should compute median
For Md
Arranging given data in ascending order
63.6, 64, 64.4, 65.3, 66.8, 67, 67.6, 67.7, 68.4, 69, 69.8,
70.2, 70.4, 71, 71.5

Here $n = 15$
let $A = x_i$ if $x_i < Md$
$B = x_i$ if $x_i > Md$
omit $x_i$ if $x_i = Md$
Then we have,
A BBB ABAAB AAABB

Now,
$Md = \left(\frac{n+1}{2}\right)^{th}$ value
$= \left(\frac{15+1}{2}\right)^{th}$ value
$= 8^{th}$ value
$= 67.7$

Here $n_1 = 7$ (no. of A), no. of Runs $(r) = 8$
$n_2 = 7$ (no. of B)
Since, level of sig $(\alpha) = 1\% = 0.01$ $(\neq 5\%)$

Here, we setup hypothesis as
$H_0$: Height of players is in random order
vs $H_1$: " " " is not in " "

Under $H_0$, test statistics,

$$z = \frac{r - \mu_r}{\sigma_r}$$

Here,
$$\mu_r = \frac{2n_1n_2}{n_1+n_2} + 1 = \frac{2\times7\times7}{7+7} + 1 = 8$$

$$\sigma_r^2 = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1+n_2)^2(n_1+n_2-1)} = \frac{2\times7\times7(2\times7\times7-7-7)}{(7+7)^2(7+7-1)}$$

$\sigma_r^2 = 3.2307$
$\sigma_r = 1.797$

Now,
$$z_{cal} = \frac{r-\mu_r}{\sigma_r} = \frac{8-8}{1.797} = 0$$

Table or critical value

$z_{tab} = 2.58$

<u>Decision:</u>

Since, $z_{cal} < z_{tab}$; we accept $H_0$ & reject $H_1$ with the conclusion that height of players is in random order

* <u>Binomial Test:-</u>

→ It is a non parametric test used to test whether the two distinct group has come from same binomial pop$^n$ or not.

For small sample $(n \leq 25)$

* <u>Test - statistics</u>

$$x_0 = \min\{n_1, n_2\}$$

where,

$n_1$ = no. of data in 1st group

$n_2$ = no. of " " 2nd "

* <u>Critical value</u>

$$p = \text{prob}\{X \leq x_0\} = \sum_{x=0}^{x_0} {}^nC_x \, p^x q^{n-x}$$

* <u>Decision:-</u>

<u>for one-tailed test:</u>

* If $p < \alpha$, we reject $H_0$ & accept $H_1$
* If $p \geq \alpha$, we accept $H_0$ & reject $H_1$

* For two-tailed test:-

* If $2p < \alpha$, we ~~need~~ reject $H_0$ & accept $H_1$

* If $2p \geq \alpha$, we accept $H_0$ & reject $H_1$

\# for large sample ($n > 25$)

Test statistics

$$\cancel{Z = \frac{X_0 - \mu}{\sigma} = X_0 - n}$$

$$Z = \frac{(x_0 \pm 0.5) - np}{\sqrt{npq}}$$

Here, we should use +ve sign if $x_0 < np$ and use -ve sign of $X_0 > np$

Q14)

Here we have,

DNDDN DND NND NNDDNN DNDD NNNN

$n_1$ = no. of Defective

$= 12 - 1 = 11$

$n_2$ = no. of non defective

$= 14$

$n = 25$

Here we setup hypothesis as

$H_0$: Def. & non def. are equally produced

vs $H_1$: " " " " " not " "

Under $H_0$ test statistics

$$X_0 = \min\{n_1, n_2\} = \min\{11, 14\}$$

$\therefore X_0 = 11$

$$p = \frac{1}{2}$$

$$q = 1 - p$$

$$= 1 - \frac{1}{2}$$

$$= 1/2$$

Now,

$$p\text{-values} = P(X \le x_0) = \sum_{x=0}^{x_0} {}^{n}C_x \, p^x q^{n-x}$$

$$= \sum_{n=0}^{11} {}^{25}C_x \left(\frac{1}{2}\right)^{x} \left(\frac{1}{2}\right)^{25-x}$$

$$= \sum_{x=0}^{11} {}^{25}C_x \left(\frac{1}{2}\right)^{25}$$

$$= \left(\frac{1}{2}\right)^{25} \left( {}^{25}C_0 + {}^{25}C_1 + {}^{25}C_2 + \cdots {}^{25}C_{11} \right)$$

$\therefore$ p-values $= 0.345$

For two tailed $2p = 2 \times 0.345 = 0.69$

Here, level of significance $(\alpha) = 5\% = 0.05$

Decision:-

Since $2p > \alpha$, we acept $H_0$ & reject $H_1$ with the conclusion that defective & non defective are equally produced.

15.) Here, we have,

Sample size $(n) = 100$

$n_1$ = no. of people who likes NOKIA

$= 40$

$n_2$ = no. of people who likes OPPO

$= 60$

$p = 1/2$

$q = 1 - 1/2 = 1/2$

Here,

we setup hypothesis as

$H_0$: NOKIA and OPPO are equally popular.

$H_1$: NOKIA and OPPO are not equally popular

Under $H_0$ test statistics

$X_0 = \min\{40, 60\}$

$= 40$

$np = 100 \times 1/2$

$= 50$

$npq = 50 \times 1/2$

$= 25$

$$z = \frac{(X_0 \pm 0.5) - np}{\sqrt{npq}}$$

Since,

$X_0 < np$, so, test statistics

$$z = \frac{(x_0 \pm 0.5) - np}{\sqrt{npq}}$$

$$= \frac{40 + 0.5 - 50}{\sqrt{25}}$$

$$= -1.9$$

$|z_{cal}| = |-1.9|$
$= 1.9$

Here,
level of significance $(\alpha) = 5\%$
$= 0.05$

For two tailed test
From SNT,
$z_{tab} = 1.96$

Decision:-
Since $z_{cal} < z_{tab}$ we accept $H_0$ & reject $H_1$ with the conclusion NOKIA and oppo are equally popular.

* Kolmogorov - Smirnov Test
It is a non-parametric test used to determine whether observed data fits certain distribution or not.

Date ____ Page ____

Test statistics

$D = Max. |F_o - F_e|$ where, $F_o = \frac{Cf_o}{N}$

$Cf_o$ = cummulative observed freq $F_e = \frac{Cf_e}{N}$

$Cf_e$ " expected "

N = Total frequency

Decision:

* If $D < D_{n,\alpha}$ we acept $H_0$ & reject $H_1$
* If $D > D_{n,\alpha}$ " reject " " acept "

where $D_{n,\alpha}$ is table value taken from K-S table on the basis of n & $\alpha$.

18)

Soln:-

Here we setup hypothesis as

$H_0$: All matched set are uniform

v/s $H_1$: " " " " not uniform

Under $H_0$, test statistics

$D = Max. |F_o - F_e|$

Calcn table for D

| M.S. | $f_o$ | $cf_o$ | $F_o$ | $f_e$ | $cf_e$ | $F_e$ | $\lvert F_o - F_e \rvert$ |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0.05 | 4 | 4 | 0.2 | 0.15 |
| 1 | 0 | 1 | 0.05 | 4 | 8 | 0.4 | 0.35 |
| 2 | 5 | 6 | 0.3 | 4 | 12 | 0.6 | 0.3 |
| 3 | 7 | 13 | 0.65 | 4 | 16 | 0.8 | 0.15 |
| 4 | 7 | 20 | 1 | 4 | 20 | 1 | 0 |
| | $N=20$ | | | $N=20$ | | | |

Here, $f_e = \frac{N}{n} = \frac{20}{5} = 4$

Here $D = 0.35$
level of significance $(\alpha) = 5\%$

<u>For two-tailed test</u>

$D_{20,\,0.05} = 0.284$

<u>Decision:-</u>

Since $D > D_{n,\alpha}$, we reject $H_0$ & acept $H_1$ with the conclusion all matched set are not uniform.

* <u>MEDIAN</u>

It is non-parametric test used to determine whether two independent sample are significantly different or not.

For small sample ($n_1 \leq 10$, $n_2 \leq 10$)
Test statistics

$$P = P(A \geq a)$$

where $P(A=a) = \dfrac{C(n_1, a) * C(n_2, K-a)}{C(n, K)}$

$n_1$ = no. of data in 1st sample
$n_2$ = no. of data in 2nd sample
$n = n_1 + n_2$
$K = \dfrac{n}{2}$

$a$ = No. of $x_i$ st $x_i \leq md$
st → such that

Now,
p-value = p, for one tailed test
p-value = 2p, for two tailed test

Decision :-
* If p-value > $\alpha$, we accept $H_0$ & reject $H_1$
* If p-value $\leq \alpha$, we reject $H_0$ & accept $H_1$

For large sample ($n_1 > 10$ or $n_2 > 10$)
First we should prepare 2x2 contingency table as

| From Sample | No. of $x_i \leq M_d$ | No. of $x_i > M_d$ | Total |
|---|---|---|---|
| 1st | a | c | a+c |
| 2nd | b | d | b+d |
| Total | a+b | c+d | N |

N=a+b+c+d

$\chi^2$ = Chi-square

Test statistics

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

If any cell frequency is less than 5 then

$$\chi^2 = \frac{N\left[|ad-bc| - \frac{N}{2}\right]^2}{(a+b)(c+d)(a+c)(b+d)}$$

Decision:-

* If $\chi^2_{cal} \le \chi^2_{(1)}$, we accept $H_0$ & reject $H_1$
* If $\chi^2_{cal} > \chi^2_{(1)}$, we reject $H_0$ & accept $H_1$

7) Soln:-

Here we setup hypothesis as

$H_0$: There is no sig. difference in two training

v/s $H_1$: There is sig difference in two training

Under $H_0$, test statistics

$p = p(A \ge a)$

where $p(A=a) = \frac{C(n_1, a) \times C(n_2, k-a)}{C(n, k)}$

Here, $n_1 = 10$, $n_2 = 10$, $k = \frac{n_1 + n_2}{2} = 10$

$n = n_1 + n_2 = 20$

For Md

Arranging given data in ascending order

30, 32, 34, 39, 40, 42, 43, 44, 45, 46

47, 47, 48, 49, 50, 51, 55, 55, 59, 71

Now,

$$Md = \left(\frac{n+1}{2}\right)^{th} \text{value}$$

$$= \left(\frac{21}{2}\right)^{th} \text{value}$$

$$= 10^{th} \text{value} + 0.5(11^{th} - 10^{th})$$

$$= 46 + 0.5(47 - 46)$$

$$Md = 46.5$$

Now,

$a$ = no. of $x_i$ s.t. $x_i \le Md$

$\therefore a = 7$

Now,

$$p = P(A \geq 7) = P(A=7) + P(A=8) + P(A=9) + P(A=10)$$

$$= \frac{C(10,7) \times C(10,3) + C(10,8) \times C(10,2) + C(10,9) \times C(10,1) + C(10,10) \times C(10,0)}{C(20,10)}$$

$\therefore p = 0.08945$

For two-tailed test

p-value $= 2p = 2 \times 0.08945$

$\therefore$ p-value $= 0.1789$

Here,

level of significant $(\alpha) = 5\% = 0.05$

<u>Decision:</u>

Since P-value > $\alpha$, we acept $H_0$ & reject $H_1$ with the conclusion that there is no sig. difference in two trainings

9.

→ Soln:-

<u>For Md</u>

arranging in ascending order

33, 36, 43, 45, 48, 60, 62, 66, 73, 73, 74, 76

77, 77, 77, 78, 78, 7[illegible], 80, 82, 85, 88, 89,

91, 93,

Here $n = 24$

Now,

$$Md = \left(\frac{n+1}{2}\right)^{th} \text{value}$$

$$= \left(\frac{24+1}{2}\right)^{th} \text{value}$$

$$= 12.5^{th}$$

$$= 12^{th} \text{value} + 0.5(13^{th} - 12^{th})$$

$$= 76 + (0.5)(77 - 76)$$

$$Md = 76.5$$

Now,

2x2 contingency for given problem

| Sample | No. of | | Total |
|---|---|---|---|
| | $x_i \leq Md$ | $x_i > Md$ | |
| I | 7 | 5 | 12 |
| II | 5 | 7 | 12 |
| Total | 12 | 12 | 24 |

Date Page

Here we setup hypothesis as

Ho: There is no significant difference in marks distn of two teachers

V/S H1: There is significant difference in marks distn of two teacher

Under Ho, test statistics

$$\chi^2 = \frac{N(ad - bc)^2}{(a+b)(a+c)(c+d)(b+d)}$$

$$= \frac{24(7\times7 - 5\times5)^2}{12\times12\times12\times12}$$

$$\chi^2_{cal} = 0.667$$

level of sig ($\alpha$) = 5%

= 0.05

d.f = 1 [(r-1)(c-1)]

From table,

$$\chi^2_{tab} = 3.841$$

Decision:-

Since, $\chi^2_{cal} < \chi^2_{tab}$ we acept Ho & reject H1 with the conclusion there is no significant difference in marks distn of two teachers.

* Kolmogorov - Smirnov test (for two sample)

It is a non-parametric test whether there is sig. difference in ~~data~~ of two series of data

Test Statistics

$$D_0 = Max. |F_1 - F_2|$$

where

$$F_1 = \frac{cf_1}{N}$$

$$F_2 = \frac{cf_2}{N}$$

$cf_1$ = com. freq of 1st series of data

$cf_2$ = com. freq of 2nd series of data

1, Here we setup Hypothesis as

$H_0$: There is no sig. difference in two type of motor.

v/s $H_1$: There is sig. difference in two type of motor

Under $H_0$, test statistics,

$$D = Max. |F_1 - F_2|$$

calc^n table for D

| type | $f_1$ | $cf_1$ | $F_1$ | $f_2$ | $cf_2$ | $F_2$ | $\lvert F_1 - F_2 \rvert$ |
|---|---|---|---|---|---|---|---|
| 2 | 0 | 0 | 0 | 1 | 1 | 1/9 | 1/9 |
| 3 | 3 | 3 | 3/9 | 2 | 3 | 3/9 | 0 |
| 4 | 2 | 5 | 5/9 | 2 | 5 | 5/9 | 0 |
| 5 | 2 | 7 | 7/9 | 1 | 6 | 6/9 | 1/9 |
| 6 | 1 | 8 | 8/9 | 2 | 8 | 8/9 | 0 |
| 7 | 1 | 9 | 1 | 1 | 9 | 1 | 0 |
| | $N = 9$ | | | $N = 9$ | | | |

Here, $D = Max \lvert F_1 - F_2 \rvert$

$$D = \frac{1}{9}, \quad N = 9$$

Here, level of significance $(\alpha) = 5\%$

$= 0.05$

From K-S table

$$D_{9, 0.05} = 5/9$$

Decision:-

Since, $D < D_\alpha$, we accept $H_0$ & reject $H_1$ with the conclusion that there is no sig. difference in two type of motor

## Mann-Whitney U-test

It is a non-parametric test used to determine whether two independent sample come from same distribution or not.

For small sample ($n_1 \le 10$, $n_2 \le 10$)

Test-statistics

$$U_0 = \min \{U_1, U_2\}$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

st $U_1 + U_2 = n_1 n_2$

$R_1$ = sum of ranks of $1^{st}$ series of data

$R_2$ = " " " " $2^{nd}$ " " "

## Critical value

$P = \text{prob.}(U \le U_0)$

## Alt.

$U_{tab} = U_{\alpha}(n_1, n_2)$ for two tailed test

## Decision

- If p-value $< \alpha$, we reject $H_0$ & accept $H_1$
- If p-value $>= \alpha$, we accept $H_0$ & reject $H_1$

Here, p-value = p for one tailed test

p-value = 2p for two " "

* for large sample ($n_1 > 10$, $n_2 > 10$)

Test statistics

$$z = \frac{U_0 - \mu_\alpha}{\sigma_n}$$

where $\mu_\alpha = \frac{n_1 n_2}{2}$

$$\sigma_U^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

* If there is tie in ranks then,

$$\sigma_U^2 = \frac{n_1 n_2}{n(n-1)} \left\{ \frac{n^3 - n}{12} - \frac{\sum t_i^3 - t_i}{12} \right\}$$

14.

→ Soln:

Here, we setup hypothesis as

Ho: There is no significant difference in two types of managers.

V/S H1: There is significant difference in two types of managers.

Under Ho, test statistics,

$U_0 = \min\{U_1, U_2\}$

Here, we have

$n_1 = 5$, $n_2 = 4$

Calcn of $R_1$ & $R_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| Trained | 78 | 64 | 75 | 45 | 82 | |
| Rank | 7 | 4 | 6 | 1 | 8 | $R_1 = 26$ |
| Untrained | 110 | 70 | 53 | 51 | | |
| Rank | 9 | 5 | 3 | 2 | | $R_2 = 19$ |

Now,

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$= 5 \times 4 + \frac{5\times 6}{2} - 26$$

$$U_1 = 9$$

& $$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

$$= 5\times 4 + \frac{4(4+1)}{2} - 19$$

$$U_2 = 11$$

Now,

$$U_0 = \text{Min}\{U_1, U_2\} = \{9, 11\} = 9$$

Here

$$p = \text{prob.}\{U \le U_0\} = \text{prob}\{U \le 9\}$$

$$\therefore p = 0.4524$$

For two tailed test,

$$p\text{-value} = 2 \times 0.4524 = 0.9048$$

Here level of significance ($\alpha$) = 5% = 0.05

Decision:

since p-value > $\alpha$, we accept $H_0$ & reject $H_1$ with the conclusion there is no significant difference in two types of managers

16)

Soln:-

Here we setup hypothesis as:

$H_0$: There is no significant difference in two varieties of wheat

v/s $H_1$: There is significant difference in two varieties of wheat

Under $H_0$, test statistics

$$z = \frac{U_0 - \mu_u}{\sigma_u}$$

Calcn table for various value

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Wheat - I | 15.9 | 15.3 | 16.4 | 14.9 | 15.3 | 16.0 | 14.6 | 15.3 | 14.5 | 16.6 | 16.0 |
| Rank | 9 | 5 | 12.5 | 3 | 5 | 10.5 | 2 | 5 | 1 | 14 | 10.5 |
| Wheat - II | 16.4 | 16.8 | 17.1 | 16.9 | 18.0 | 15.6 | 18.1 | 17.2 | 15.4 | | |
| Rank | 12.5 | 15 | 17 | 16 | 19 | 8 | 20 | 18 | 7 | | |

$R_1 = 77.5$

$R_2 = 132.5$

Here, $n_1 = 11$, $n_2 = 9$, $t_1 = 3$, $t_2 = 2$, $t_3 = 2$

Now,

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$= 11 \times 9 + \frac{11 \times 12}{2} - 77.5$$

$\therefore U_1 = 87.5$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

$$= 11 \times 9 + \frac{9 \times 10}{2} - 132.5$$

$U_2 = 11.5$

Here, $U_0 = \min\{U_1, U_2\} = \min\{87.5, 11.5\}$
$= 11.5$

$$\mu_U = \frac{n_1 n_2}{2} = \frac{11 \times 9}{2} = 49.5$$

$$\sigma_U^2 = \frac{n_1 n_2}{n(n-1)} \left[ \frac{n^3 - n}{12} - \frac{\sum(t_i^3 - t_i)}{12} \right]$$

$$= \frac{99}{380} \left[ \frac{20^3 - 20}{12} - \frac{(3^3 - 3) + (2^3 - 2) + (2^3 - 2)}{12} \right]$$

16.0

10.5

$\sigma_U = 13.133$

Now,

$$Z_{cal.} = \frac{11.5 - 49.5}{13.133} = -2.89$$

$$|Z_{cal}| = 2.89$$

Here, level of sig ($\alpha$) = 5%
For two tailed test
$z_{tab} = 1.96$

Decision:
since $z_{cal} > z_{tab}$. we reject $H_0$ & accept $H_1$ with the conclusion there is significant difference in two variables of wheat.

* Chi-Square Test ($\chi^2$ - test)

Applications:-
1. Uniformity test / Randomness test
2. Goodness fit
3. Association / Independence test

* For goodness of fit

Test- statistics

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

where, O = Observed frequency
E = Expected frequency

For binomial
E = N × prob.
$\therefore E = N \times {}^nC_r \, p^r q^{n-r}$

For poisson

$$E = N \times \frac{e^{-\lambda} \lambda^{r}}{r!}$$

For Association test

Test - statistics

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

where,

$$E = \frac{RT \times CT}{GT}$$

RT = Row total
CT = Column total
GT = Grand total

Note:- For 2x2 contingency table, if any cell frequency is less than 5, we should use yate's correction for the computation of $\chi^2$ as,

$$\chi^2 = \frac{N\left[|ad - bc| - \frac{N}{2}\right]^2}{(a+b)(a+c)(b+d)(c+d)}$$

Here,

d.f = (r-1)(c-1)

r = No. of rows
c = No. of columns

| | |
|---|---|
| a | b |
| c | d |

| | |
|---|---|
| a | c |
| b | d |

19.

→ Here, we setup hypothesis as

$H_0$: Die is fair

v/s $H_1$: Die is biased

Under $H_0$ test statistics:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Here,

O = observed freq.

E = Expected freq $= \frac{\sum O}{n}$

Here n = 6

Calcn table for $\chi^2$ statistics

| O | E | $(O-E)^2/E$ |
|---|---|---|
| 8 | 10 | 0.4 |
| 9 | 10 | 0.1 |
| 13 | 10 | 0.9 |
| 7 | 10 | 0.9 |
| 15 | 10 | 2.5 |
| 8 | 10 | 0.4 |
| $\sum O = 60$ | $\sum E = 60$ | $\sum \frac{(O-E)^2}{E} = 5.2$ |

Here, $E = \frac{\sum O}{n} = \frac{60}{6} = 10$

Here, $\chi^2_{cal} = 5.2$

level of sig $(\alpha)$ = 5%
= 0.05

d.f = n − 1 = 6 − 1 = 5

From table,

$f^2_{tab}$ = 11.07

Decision:

Since $f^2_{cal} < f^2_{tab}$, we accept $H_0$ & reject $H_1$ with the conclusion that Die is fair.

Q0
→

Hint:- $E(SC) = \frac{62}{62+4+64} \times 200 = 124$

$E(M) = \frac{4}{100} \times 200 = 8$

$E(O) = \frac{34}{100} \times 200 \times 68$

22

→ Soln:

Here, we setup Hypothesis as,

$H_0$: Data fit genetic theory

v/s $H_1$: Data doesn't fit genetic theory.

Under $H_0$, test statistics,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed frequency

E = Expected frequency $= \frac{\sum O}{n}$

n = 3

Calch table for $\chi^2$-statistics

| Type | O | E | $(O-E)^2/E$ |
|---|---|---|---|
| A | 90 | 75 | 3 |
| AB | 135 | 150 | 1.5 |
| B | 75 | 75 | 0 |
| | $\Sigma O = 300$ | | $\Sigma \frac{(O-E)^2}{E} = 4.5$ |

Here, $E = \frac{\Sigma O}{n} = \frac{300}{3} = 100$

Here $\chi^2_{cal} = 4.5$

level of signi $(\alpha) = 5\%$
$= 0.05$

$df = n-1$
$= 3-1$
$= 2$

Here $\chi^2_{tab} = 5.991$

Decision:-

Since $\chi^2_{cal} < \chi^2_{tab}$ we accept $H_0$ & reject $H_1$ with the conclusion that data fit genetic theory.

Note:-

For fitting Binomial and Poisson $dist^n$.

$d.f = n-1-k_1-k_2$

$n$ = no. of data

$k_1$ = no. of parameter estimated

$k_2$ = no. of expected frequency less than 5

23)

→ Soln:-

Since $\lambda$ (mean) of the $dist^n$ is not given, we estimate it;

$$\lambda = \bar{x} = \frac{\Sigma fx}{N} \quad [x = \text{No. of misprints}]$$

$$= \frac{221\times0 + 167\times1 + 70\times2 + 30\times3 + 7\times4 + 5\times5}{221 + 167 + 70 + 30 + 7 + 5}$$

$$\lambda = 0.9 \quad , N = 500$$

Now, expected freq. (E) = $N \times$ prob.

$$= N \times \frac{e^{-\lambda} \lambda^{r}}{r!}$$

$$\therefore E = 500 \times e^{-0.9} \times \frac{0.9^r}{r!}$$

for $r = 0, 1, 2, 3, 4, 5$

$E(0) = 203.28$

$E(1) = 182.95$

$E(2) = 82.33$
$E(3) = 24.~~687~~$
$E(4) = 5.56$
$E(5) = 1.00$

We setup hypothesis as
$H_0$: Data fits poisson distn
v/s $H_1$: Data does not fit Poisson distn

Under $H_0$, test statistics

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Now
calcn table for $\chi^2$ - statistics

| $X=r$ | O | E | $(O-E)^2/E$ |
|---|---|---|---|
| 0 | 221 | 203.28 | 1.545 |
| 1 | 167 | 182.95 | 1.391 |
| 2 | 70 | 82.33 | 1.847 |
| 3 | 30 | 24.7 | 1.137 |
| 4 | 7 } 12 | 5.567 } 6.556 | |
| 5 | 5 | 1 | 4.511 |
| | $\sum O = 500$ | $\sum E = 500$ | $\sum \frac{(O-E)^2}{E} = 10.431$ |

Here
level of significance $(\alpha) = 5\% = 0.05$
$df = n - 1 - k_1 - k_2$
$= 6 - 1 - 1 - 1$
$\therefore d.f = 3$

From table

$\therefore \chi^2_{tab} = 7.815$

Decision:-

since $\chi^2_{cal} > \chi^2_{tab}$ we reject $H_0$ & accept $H_1$ with the conclusion that data does not fit poisson distn.

24)

→ Soln:-

Here we setup hypothesis as

$H_0$: Binomial distn fits on given data

v/s $H_1$: Binomial distn doesnot fits on given data

Under $H_0$, test statistics,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Here,

$E = N \times prob = N \times {}^nC_r \cdot p^r \cdot p^{n-r}$

Here, $N = 50$

$n = 6$

$p = 0.30$

$q = 0.7$

Now

$$E = 50 \times {}^5C_r \times 0.3^r \times 0.7^{5-r}$$

for $r = 0, 1, 2, 3, 4, 5$

$E(0) = 84$
$E(1) = 18.01$
$E(2) = 15.44$
$E(3) = 6.62$
$E(4) = 1.42$
$E(5) = 0.12$

Now,
calcn table for $\chi^2$ statistics

| $x = r$ | $O$ | $E$ | $(O-E)^2/E$ |
|---|---|---|---|
| 0 | 4 | 84 | 2.305 |
| 1 | 13 | 18.01 | 1.394 |
| 2 | 17 | 15.44 | 0.158 |
| 3 | 12 | 6.62 | |
| 4 | 3 | 1.42 | 7.561 |
| 5 | 1 | 0.12 | |
| | $\Sigma O = 50$ | $\Sigma E = 50$ | $\Sigma \frac{(O-E)^2}{E} = 11.4$ |

Here,

$$\chi^2_{cal} = 11.42$$

level of sig $(\alpha) = 5\%$

$$d.f = n - 1 - k_1 - k_2 = 6 - 1 - 0 - 2 = 3$$

From table,
Here
$\chi^2_{tab} = 7.815$

Decision:-
Since $\chi^2_{cal} > \chi^2_{tab}$ we reject $H_0$ &
accept $H_1$ with the conclusion that the binomial distn doesnot fits on given data

Q8) Soln:-
Here we setup hypothesis as,
$H_0$: There is no association between father's & Son's eye color.
V/S $H_1$: There is associative between Father's & Son's eye color.
Under $H_0$, test statistics,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

where, $E = \frac{RT \times CT}{GT}$

Here,
RT = Row Total
CT = Column Total
GT = Grand Total

Here, we have

| Father's eye color | Son's eye color NL | L | Row Total |
|---|---|---|---|
| NL | 230 | 148 | 378 |
| L | 151 | 471 | 622 |
| Col Total | 381 | 619 | 1000 = GT |

Here,

$E(230) = \frac{378 \times 381}{1000} = 144.02$

$E(148) = \frac{378 \times 619}{1000} = 233.98$

$E(151) = \frac{622 \times 381}{1000} = 236.98$

$E(471) = \frac{622 \times 619}{1000} = 385.02$

Now,

Calcⁿ table for $\chi^2$ - statistics

| O | E | $(O-E)^2/E$ |
|---|---|---|
| 230 | 144.02 | 51.33 |
| 148 | 233.98 | 31.59 |
| 151 | 236.98 | 31.19 |
| 471 | 385.02 | 19.20 |
| $\Sigma O = 1000$ | $\Sigma E = 1000$ | $\Sigma \frac{(O-E)^2}{E} = 133.13$ |

Here,

$\chi^2_{cal} = 133.13$

level of sig $(\alpha) = 5\% = 0.05$

$df = (r-1)(c-1) = (2-1)(2-1)$

$df = 1$

From table,

$\chi^2_{tab} = 3.841$

Decision:

Since $\chi^2_{cal} > \chi^2_{tab}$ we reject $H_0$ and accept $H_1$ with the conclusion that there is association betn Father's & Son's eye color.

2g) Soln:-

Here we setup hypothesis as,

$H_0$: There is no association betn sex & smoking habit

v/s $H_1$:- There is association betn sex and smoking habit

Under $H_0$, test statistics

$$\chi^2 = \frac{N\left[|ad-bc| - \frac{N}{2}\right]^2}{(a+b)(a+c)(c+d)(b+d)}$$ [since one cell value is less than 5]

Here, we have

| Habit | Sex M | F | Total |
|---|---|---|---|
| S | 40 | 33 | 73 |
| N | 3 | 12 | 15 |
| Total | 43 | 45 | 88 |

| | |
|---|---|
| a | b |
| c | d |

Here,

$$\chi^2_{cal} = \frac{88\left[|40\times 12 - 33\times 3| - \frac{88}{2}\right]^2}{(40+33)(40+3)(3+12)(33+12)}$$

$$\chi^2_{cal} = 4.72$$

Here,

level of sig($\alpha$) = 5% = 0.05

df = (r−1)(c−1) = (2−1)(2−1)

= 1

from table

$\chi^2_{tab}$ = 3.841

Decision:-

Since, $\chi^2_{cal} > \chi^2_{tab}$ we reject $H_0$ & accept $H_1$ with the conclusion that there is association betn sex & smoking habit.

3) Soln:-

Here we setup hypothesis as:

$H_0$: There is no association betn new drug and attack of influnza.

$H_1$: There is association betn new drug and attack of influnza.

Under $H_0$, test statistics,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Here,

$$E = \frac{RT\times CT}{GT}$$

Here, we have

| Attack Inf. | New Drug A | $\bar{A}$ | Row Total |
|---|---|---|---|
| Y | 24 | 32 | 56 |
| N | 52 | 12 | 64 |
| Col. Total | 76 | 44 | 120 = GT |

Here,

$$E(24) = \frac{56 \times 76}{120} = 35.47$$

$$E(32) = \frac{56 \times 44}{120} = 20.53$$

$$E(52) = \frac{64 \times 76}{120} = 40.53$$

$$E(12) = \frac{64 \times 44}{120} = 23.47$$

Now,

Calc$^n$ table for $\chi^2$ - statistics

| O | E | $(O-E)^2/E$ |
|---|---|---|
| 24 | 35.47 | 3.71 |
| 32 | 20.53 | 6.41 |
| 52 | 40.53 | 3.25 |
| 12 | 23.47 | 5.61 |
| $\Sigma O = 120$ | $\Sigma E = 120$ | |

$$\Sigma \frac{(O-E)^2}{E} = 18.98$$

Here

$\chi^2_{cal} = 18.98$

level of signi($\alpha$) = 5%
= 0.05

df = (r-1)(c-1)
= 1

From tab,

$\chi^2_{tab} = 3.841$

<u>Decision:</u>

Since $\chi^2_{cal} > \chi^2_{tab}$ we reject $H_0$ &
acept $H_1$ with the conclusion that there
is association betn new drug & attack of
influnza.

## Wilcoxon Matched pair signed rank test

* For small sample ($n \leq 25$)

Test - Statistics

$$T = \min \{ S(+), S(-) \}$$

where, $S(+)$ = sum of positive signed rank
$S(-)$ = sum of negative signed rank

Table value

$T_{n,\alpha}$ is observed from table for n & $\alpha$
n = Sample size after omitting $d_i = 0$

Decision:-

* If $T \leq T_{n,\alpha}$, we reject $H_0$ & accept $H_1$
* If $T > T_{n,\alpha}$, we accept $H_0$ & reject $H_1$

* For large sample ($n > 25$)

Test - statistics

$$Z = \frac{T - \mu_T}{\sigma_T}$$

Where,

$$\mu_T = \frac{n(n+1)}{4}$$

$$\sigma_T^2 = \frac{n(n+1)(2n+1)}{24}$$

7)

Soln:

Here we setup hypothesis as,

$H_0$: Smoking has no effect on person's weight

v/s $H_1$: Smoking has effect on person's weight

Under $H_0$, test statistics

$T = \min\{S(+), S(-)\}$

Calcn for $S(+)$, $S(-)$

| X | Y | d | Rank |
|---|---|---|---|
| 66 | 71 | -5 | -5 |
| 80 | 82 | -2 | -2.5 |
| 69 | 68 | 1 | 1 |
| 52 | 56 | -4 | -4 |
| 75 | 73 | 2 | 2.5 |

Here,

$d = x - y$

$S(+) = 3.5 \ (1 + 2.5)$

$S(-) = 11.5 \ (-5 - 4 - 2.5)$

Now,

$T = \min\{S(+), S(-)\}$

$T = \min\{3.5, 11.5)$

$\therefore T = 3.5$

Here,

level of sig($\alpha$) = 5% = 0.05, n = 5

From tables (for OTT)

$T_{n,\alpha} = 1$

Decision:-

Since $T > T_{n,\alpha}$ we accept $H_0$ & reject $H_1$ with conclusion that smoking has no ~~e~~effect on person's weight

Q 10)

Soln:-

Here we setup hypothesis as

$H_0$: There is no. sig difference between X & Y

$H_1$: ~~There is sig~~ Arch. X is batter than Arch. Y.

~~Under $H_0$, test statistics~~

Calch tab for S(+), S(-)

| X | Y | d | Rank |
|---|---|---|---|
| 1 | 0 | 1 | 7 |
| 0 | 0 | – | – |
| 2 | 1 | 1 | 7 |
| 3 | 0 | 3 | 22 |
| 1 | 2 | -1 | -7 |
| 0 | 0 | – | – |
| 2 | 0 | 2 | 17 |
| 2 | 1 | 1 | 7 |
| 3 | 1 | 2 | 17 |
| 0 | 2 | -2 | -17 |
| 1 | 0 | 1 | 7 |

| | | | |
|---|---|---|---|
| 1 | 1 | – | – |
| 4 | 2 | 2 | 17 |
| 1 | 1 | – | – |
| 2 | 1 | 1 | 7 |
| 1 | 0 | 1 | 7 |
| 3 | 2 | 1 | 7 |
| 5 | 2 | 3 | 22 |
| 2 | 6 | -4 | -25 |
| 1 | 0 | 1 | 7 |
| 3 | 2 | 1 | 7 |
| 2 | 3 | -1 | -7 |
| 4 | 0 | 4 | 25 |
| 1 | 2 | -1 | -7 |
| 3 | 1 | 2 | 17 |
| 2 | 0 | 2 | 17 |
| 0 | 1 | -1 | -7 |
| 2 | 0 | 2 | 17 |
| 4 | 1 | 3 | 22 |
| 1 | 5 | -4 | -25 |

Now,

$S(+) = 7 \times 9 + 22 \times 3 + 17 \times 6 + 25$

$= 256$

$S(-) = (-7) \times 4 + (-17) \times 1 + (-25) \times 2$

$= 95$

Here, $T = \min\{S(+), S(-)\}$

$= \min\{256, 95\}$

$= 95$

Here,

$$n = 30 - 4$$
$$= 26$$

Now,

$$\mu_T = \frac{n(n+1)}{4}$$
$$= \frac{26 \times 27}{4}$$
$$= 175.5$$

$$\sigma_T^2 = \frac{n(n+1)(2n+1)}{24}$$
$$= \frac{26(27)(26 \times 2 + 1)}{24}$$
$$\sigma_T = 36.37$$

For large sample

Test statistics,

$$z = \frac{T - \mu_T}{\sigma_T}$$
$$= \frac{95 - 175.5}{36.37}$$
$$|z_{cal}| = |-2.21|$$

$$z_{cal} = 2.21$$

Here

level of sig ($\alpha$) = 5%

= 0.05

For one tailed test,

$Z_{tab} = 1.65$

Decision:

Since $Z_{cal} > Z_{tab}$, we reject $H_0$ & accept $H_1$ with the conclusion that Arch x is better than Arch y.

9)

→ Soln:-

Here we setup hypothesis as

$H_0$:

Date
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* Cochran - Q Test

Test - statistics

$$Q = \frac{(K-1)\left[K \sum Ri^2 - (\sum Ri)^2\right]}{K \sum Cj - \sum Cj^2}$$

where,

$K$ = no. of treatments
$r$ = row value
$c$ = column value

Table value :-
we should take $\chi^2_{\alpha,(K-1)}$ as table value.

Decision:-
* If $Q \leq \chi^2_{\alpha,(K-1)}$, we accept $H_0$ & reject $H_1$
* If $Q > \chi^2_{\alpha,(K-1)}$, we reject $H_0$ & accept $H_1$

11) Soln:-
Here we setup hypothesis as
$H_0$: There is no significant difference in 4 objective questions
V/S
$H_1$: There is significant difference in four objective question
Under $H_0$, test statistics,

$$Q = (K-1)\frac{\left[K \sum Ri^2 - (\sum Ri)^2\right]}{K \sum Cj - \sum Cj^2}$$

Calculation table for Q

| | Students 1 | 2 | 3 | 4 | 5 | $R_i$ | $R_i^2$ |
|---|---|---|---|---|---|---|---|
| $Q_1$ | 1 | 0 | 0 | 1 | 1 | 3 | 9 |
| $Q_2$ | 0 | 1 | 1 | 0 | 0 | 3 | 9 |
| $Q_3$ | 1 | 1 | 1 | 0 | 1 | 3 | 9 |
| $Q_4$ | 0 | 0 | 1 | 0 | 1 | 2 | 4 |
| $C_j$ | 2 | 2 | 3 | 1 | 3 | $\sum R_i = \sum C_j = 11$ | $\sum R_i^2 = 31$ |
| $C_j^2$ | 4 | 4 | 9 | 1 | 9 | $\sum C_j^2 = 27$ | |

Here, $k = 4$

Now,

$$Q = \frac{(4-1)(4 \times 31 - 11^2)}{4 \times 11 - 27}$$

$= 0.529$

Here,

level of significance $(\alpha) = 5\%$
$= 0.05$

$df = k-1 = 4-1 = 3$

From table,

$$\chi^2_{\alpha,(k-1)} = 7.815$$

Decision:

Since $Q < \chi^2_{\alpha,(k-1)}$, we accept $H_0$ & reject $H_1$ with the conclusion that there is no significance difference in for O.Q.

Date ____ Page ____

* Kruskal - Wallis H-Test

-> It is used to test the significant diff in three or more than three independent $pop^n$. It is also known as one way ANOVA

* Test-statistics

$$H = \frac{12}{n(n+1)} \sum R_i^2/n_i - 3(n+1)$$

If there is repitition in rank, then

$$H = \frac{\frac{12}{n(n+1)} \sum R_i^2/n_i - 3(n+1)}{1 - \frac{\sum (t_i^3 - t_i)}{n(n^2-1)}}$$

* Critical value:

* For $n_i \le 5$ & $k=3$, value of 'p' is taken for K-W test.

* For $n_i > 5$ & $k > 3$, critical value is $\chi^2_{\alpha, (k-1)}$.

14) Soln:-

Here we setup hypothesis as,

$H_0$: All fertilizers are equally effective

vs $H_1$: All fertilizers are not equally effective

Under $H_0$, test statistics

$$H = \frac{\frac{12}{n(n+1)} \sum R_i^2/n_i - 3(n+1)}{1 - \frac{\sum(t_i^3 - t_i)}{n(n^2-1)}}$$

[For repititative rank case]

calculation of H

| | | | | |
|---|---|---|---|---|
| N | 122 | 80 | 138 | 121 |
| Rank | 11 | 6 | 12 | 10 |
| P | 81 | 80 | 79 | 65 |
| Rank | 8 | 6 | 4 | 2.5 |
| K | 80 | 82 | 65 | 58 |
| Rank | 6 | 9 | 2.5 | 1 |

Here, $n = 12$, $t_1 = 2$, $t_2 = 3$

$n_1 = 4$, $n_2 = 4$, $n_3 = 4$ $\sum R_i^2/n_i = 5070.87$

Now,

$$H = \frac{\frac{12}{n(n+1)} \sum R_i^2/n_i - 3(n+1)}{1 - \frac{(2^3-2)+(3^3-3)}{12(12^2-1)}}$$

$$= \frac{\frac{12}{12(12+1)} \times 5070.87 - 3(12+1)}{1 - \frac{(2^3-2)+(3^3-3)}{12(12^2-1)}}$$

$\frac{4.914}{0.983}$

$= 5$

Here, $H = 5$, $k-1 = 3-1 = 2$

Here, level of $sig(\alpha) = 5\% = 0.05$
From K-W table,

$p = 0.054$

**Decision :-**
Since p-value > $\alpha$, we reject $H_1$ & accept $H_0$ with conclusion that all fertilizers are equally effective.

$R_i^2/n_i$

380.25

105.06

85.56

13
→ Soln:-

Here we setup hypothesis as,
$H_0$: All three methods are equally effective
vs $H_1$: All three methods are not equally effective

Under $H_0$, test statistics

$$H = \frac{\frac{12}{n(n+1)} \sum \left(\frac{R_i^2}{n_i}\right) - 3(n+1)}{1 - \frac{\sum (t_i^3 - t_i)}{n(n^2-1)}}$$

[For repitative rank case]

Calc$^h$ of H

*Method I

* ~~Fri~~edmann's F-test:-

It is non-parametric test used to determine sig difference in ~~2 or than~~ 3 or more than ~~two~~ 3 independent pop$^n$. It is also known as two way ANOVA.

Test-statistics

$$F_r = \frac{12}{nK(K+1)} \sum R_i^2 = 3n(K+1)$$

If tie in rank occurs

$$F_r = \frac{\frac{12}{nk(k+1)} \sum R_i^2 - 3n(k+1)}{1 - \frac{\sum (t_i^3 - t_i)}{n(k^3 - k)}}$$

Critical value :-

For $2 \le n \le 9$ & $k = 3$ / $2 \le n \le 4$ & $k = 4$
p-value is obtained from Friedman's table

For $n \geq 5$ & $k > 3$

Critical value is $\chi^2_{\alpha, (k-1)}$

Q 18

Here we have setup hypothesis as,

$H_0$ : There is no significant difference in three Eng. school.

v/s $H_1$ : There is significant difference in three Eng school

Calc table for F - statistics,

| School | Grade I | Rank | II | Rank | III | Rank | IV | Rank | R |
|---|---|---|---|---|---|---|---|---|---|
| Alpha | 89 | 3 | 98 | 3 | 70 | 3 | 80 | 3 | 12 |
| Sigma | 45 | 2 | 76 | 2 | 40 | 2 | 55 | 1 | 7 |
| Gamma | 20 | 1 | 58 | 1 | 35 | 1 | 67 | 2 | 5 |

Here, no. of school (k) = 3
no. of grade (n) = 4

Now,
Under $H_0$, Test-statistics

$$F_r = \frac{12}{nk(k+1)} \sum R_i^2 - 3n(k+1)$$

$$= \frac{12}{4 \times 3(4)} 218 - 3 \times 4(4)$$

$\therefore F = 6.5$

Here,
level of sig ($\alpha$) = 50%
= 0.05

For n=4, k=3, from Friedman's table

| Rank | R | $R^2$ |
|---|---|---|
| | 12 | 144 |
| | 7 | 49 |
| | 5 | 25 |

$\Sigma R_i^2 = 218$

p-value = 0.042

Decision:-

Since, p-value < $\alpha$ we reject $H_0$ & accept $H_1$ with the conclusion that there is significant difference in three Eng. school.

19.

→ Here we setup hypothesis as ~~that the~~

$H_0$: There is no significant birth rate ~~is~~ over all four seasons

v/s $H_1$: There is significant difference in birth rate over all four season.

Calcn table for F - statistics

| $R_i^2$ | R | Hospital | A | Rank | B | Rank | C | Rank | D | Rank |
|---|---|---|---|---|---|---|---|---|---|---|
| 56.25 | 7.5 | Winter | 92 | 2 | 9 | 1 | 58 | 2 | 19 | 2.5 |
| 225 | 15 | Spring | 112 | 4 | 11 | 3 | 71 | 4 | 26 | 4 |
| 72.25 | 8.5 | Summer | 94 | 3 | 10 | 2 | 51 | 1 | 19 | 2.5 |
| 81 | 9 | Fall | 77 | 1 | 15 | 4 | 62 | 3 | 18 | 1 |

~~$R_1$ = 7.5~~ $\Sigma R_i^2 = 434.5$

~~$R_i^2$ =~~

Here
no. of hospital $(k) = 4$
no. of ~~birth~~ birth $(n) = 4$

Now,
Under $H_0$, test statistics,

$$F_r = \frac{\frac{12}{nk(k+1)} \sum R_i^2 - 3n(k+1)}{1 - \frac{\sum(t_i^3 - t_i)}{n(k^3 - k)}}$$

$t_1 = 2$

$$= \frac{\frac{12}{4 \times 4(5)} \, 434.5 - 3 \times 4(4+1)}{1 - \frac{2^3 - 2}{4(4^3 - 4)}}$$

$F_r = 57.62$

Here

Completely Randomized Design C

→ It is the simplest design based on only two principle of expiremental design namely Viz. Replication & Randomization. Here treatments are allocated in expiremental unit with completely randomized way. Here treatments are replicated in fair number such that error sum of squares will be minimized and precision of the design will be maximized. Here we use one way Anova to analyze the design.

* Layout of CRD

let us consider three treatments then layout of CRD maybe

| | | | |
|---|---|---|---|
| A | B | C | C |
| A | C | B | A |
| B | A | C | B |

* Statistical model of CRD

Let us consider 't' treatment replicated 'r' times, then Statistical model of CRD is.

$$y_{ij} = \mu + t_i + e_{ij} \quad \text{— (e)}$$

Here,

$i = 1, 2, ---, t$

& $j = 1, 2, ---, r$

iid → identically independent distribution

where, $y_{ij}$ = yield or observation from $i^{th}$ treatment & $j^{th}$ replication

$\mu$ = overall mean

$\tau_i$ = $i^{th}$ treatment effect

$\epsilon_{ij}$ = Error or residual term such that $\epsilon_{ij}$ are ~~iid~~ r.v with $\epsilon_{ij} \sim N(0, \sigma_e^2)$

* <u>Statistical analysis of CRD</u> :-

There, we partition total sum of square into its component part as

$$\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{..})^2 = \sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i.}+\bar{y}_{i.}-\bar{y}_{..})^2$$

where,

$$\bar{y}_{..} = \frac{\sum_{i=1}^{t}\sum_{j=1}^{r} y_{ij}}{t.r} = \frac{y_{..}}{t.r} = \text{overall mean}$$

$$\bar{y}_{i.} = \frac{\sum_{j=1}^{r} y_{ij}}{r} = \frac{y_{i.}}{r} = i^{th}\text{ treatment mean}$$

$$= \sum_{i=1}^{t}\sum_{j=1}^{r}\{(\bar{y}_{i.}-\bar{y}_{..})+(y_{ij}-\bar{y}_{i.})\}^2$$

$$= \sum_{i=1}^{t}\sum_{j=1}^{r}(\bar{y}_{i.}-\bar{y}_{..})^2+\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i.})^2 + \text{Cross product term}$$

$$= r\sum_{i=1}^{t}(\bar{y}_{i.}-\bar{y}_{..})^2+\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i.})^2$$

[Cross product term vani~~able~~shes as sum of deviation taken from A.M is zero]

$\boxed{TSS = SST + SSE}$

Here,
$TSS = \sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij} - \bar{y}_{..})^2$ = Total sum of square

$SST = r\sum_{i=1}^{t}(\bar{y}_{i.} - \bar{y}_{..})^2$ = Sum of square due to treatment

$SSE = TSS - SST$ = Sum of square due to error

- For Practical Purpose

$TSS = \sum\sum y_{ij}^2 - C.F.$

$SST = \frac{\sum y_{i.}^2}{r} - C.F$

$C.F$ = correction factor $= \frac{G^2}{N}$

G = Grand total
N = Total no. of observation

One-way ANOVA table for CRD

| S.V | d.f | SS | MSS | F-ratio |
|---|---|---|---|---|
| Treatment | $t-1$ | SST | $MSST = \frac{SST}{df}$ | $F = \frac{MSST}{MSSE}$ |
| Error | $t(r-1)$ | SSE | $MSSE = \frac{SSE}{df}$ | |
| Total | $t.r-1$ | TSS | | |

Decision:

* If $f_{cal.} \le f_{tab.}$, we accept $H_0$ & reject $H_1$
* If $f_{cal} > f_{tab}$ we reject $H_0$ & accept $H_1$

* Question practice

1)

→ Given design is CRD with three treatments
calcn table for CRD

| Treatments | $y_{i.}$ |
|---|---|
| A: 10 + 15 + 20 + 16 | 61 |
| B: 5 + 6 + 10 + 12 | 33 |
| C: 15 + 11 + 22 + 18 | 66 |
| | G = 160 |

Here,

$t = 3$, $r = 4$

$N = 12$

for C.F

$$C.F = \frac{G^2}{N}$$

$$= \frac{160^2}{12}$$

$$CF = 2133.33$$

For TSS

$$TSS = \sum\sum y_{ij}^2 - CF$$

$$= 10^2 + 15^2 + 20^2 + \cdots + 18^2 - 2133.33$$

$$\therefore TSS = 306.67$$

for SST

$$SST = \frac{\sum y_{i.}^2}{r} - C.F$$

$$= \frac{61^2}{4} + \frac{33^2}{4} + \frac{66^2}{4} - 2133.33$$

$$= 158.17$$

For SSE

$$SSE = TSS - SST$$

$$= 306.67 - 158.17$$

$$SSE = 148.5$$

Now

ANOVA table for CRD

| S.V | d.f | S.S | M.S.S | F_ratio |
|---|---|---|---|---|
| treatments | 3-1=2 | 158.17 | 79.085 | 4.793 |
| Error | 11-2=9 | 148.5 | 16.5 | |
| Total | 12-1=11 | 306.67 | | |

Here we set up hypothesis as,

$H_0: \mu_A = \mu_B = \mu_C$ i.e There is no significant difference in three treatments.

V/s $H_1: \mu_i \neq \mu_j$ (for at least one pair $i \neq j$) There is sig. difference in three treatments.

Under Ho, test statistics

$$F = \frac{MSST}{MSSE}$$

$\therefore F_T = 4.793$

Here,

level of significant ($\alpha$) = 5% (let)

d.f = (2, 9)

From F-table

$F_{tab} = 4.256$

Decision:-

Since $F_{cal} > F_{tab}$ we can to conclusion that we reject Ho & acept $H_1$ with the conclusion that there is significant difference in three treatments.

Randomized Block Design (RBD)

→ It is the simplest design based on all three basic principles of design of exp. (viz. Randomization, Replication & local control) Here treatments are allocated in expirmental unit in such a way that treatments occur once and only once either in rows or in columns. Here treatment are replicated in fair manner in order to increase precision of the design ~~to~~ & to reduce the sum of square due to error. This design is also known as row wise or column wise design. We use two way ANOVA to analyze (RBD)

* Layout of RBD:-

let us consider four treatments then layout of RBD may be

| | | | |
|---|---|---|---|
| A | B | C | A |
| B | A | B | C |
| C | C | D | B |
| D | D | A | D |

* Statistical model of RBD.

let us consider 't' treatments replicated 'r' times then statistical model of RBD is

$$y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij} \quad \text{---} \; (i)$$

$$i = 1, 2, \ldots, t$$
$$j = 1, 2, \ldots, r$$

where,

$y_{ij}$ = yields or observation of $i$th treatments & $j$th block

$\mu$ = overall mean

$\tau_i$ = $i^{th}$ treatment effect

$\beta_j$ = $j^{th}$ Block effect

$\epsilon_{ij}$ = Error or residual term such that $\epsilon_{ij}$ are iid r.v with $\epsilon_{ij} \sim N(0, \sigma_e^2)$

* Statistical Analysis of RBD

For the analysis we partition total sum of square into its component parts as

$$\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{..})^2=\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i.}+\bar{y}_{i.}+\bar{y}_{.j}-\bar{y}_{.j}-\bar{y}_{..})^2$$

$$=\sum_{i=1}^{t}\sum_{j=1}^{r}\{(\bar{y}_{i.}-\bar{y}_{..})+(\bar{y}_{.j}-\bar{y}_{..})+(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})\}^2$$

$$=\sum_{i=1}^{t}\sum_{j=1}^{r}(\bar{y}_{i.}-\bar{y}_{..})^2+\sum_{i=1}^{t}\sum_{j=1}^{r}(\bar{y}_{.j}-\bar{y}_{..})^2+\cancel{\sum_{i=1}^{t}\sum_{j=1}^{r}}$$

$$+\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})^2+\text{Cross product term}$$

$$=r\sum_{i=1}^{t}(\bar{y}_{i.}-\bar{y}_{..})^2+t\sum_{j=1}^{r}(\bar{y}_{.j}-\bar{y}_{..})^2$$

$$+\sum_{i=1}^{t}\sum_{j=1}^{r}(y_{ij}-\bar{y}_{i.}-\bar{y}_{.j}+\bar{y}_{..})^2$$

[Cross product term vanishes as sum of deviation taken from A.M is zero]

$$\boxed{TSS = SST + SSB + SSE}$$

For practical purpose,

$$CF = \frac{G^2}{N}$$

$$TSS = \sum_{i=1}^{t} \sum_{j=1}^{r} y_{ij}^2 - CF$$

$$SST = \sum_{i=1}^{t} \frac{y_{i\cdot}^2}{r} - C.F$$

$$SSB = \sum_{j=1}^{r} \frac{y_{\cdot j}^2}{t} - C.F$$

$$\boxed{SSE = TSS - SST - SSB}$$

ANOVA table for RBD

| SV | d.f | SS | MSS | F-ratio |
|---|---|---|---|---|
| Treatment | $t-1$ | SST | $MSST = \frac{SST}{t-1}$ | $F_T = \frac{MSST}{MSSE}$ |
| Blocks | $r-1$ | SSB | $MSSB = \frac{SSB}{r-1}$ | $F_B = \frac{MSSB}{MSSE}$ |
| Error | $(t-1)(r-1)$ | SSE | $MSSE = \frac{SSE}{(t-1)(r-1)}$ | |
| Total | $t.r-1$ | TSS | | |

Decision : As in CRD

12. Here,

Given design is RBD with 3-treatments & 4-Blocks

Now, calculation table for various sums

| Treat | Blocks I | II | III | IV | $y_{i\cdot}$ |
|---|---|---|---|---|---|
| A | 8 | 8 | 6 | 8 | 30 |
| B | 12 | 8 | 9 | 10 | 37 |
| C | 10 | 10 | 10 | 9 | 41 |
| $y_{\cdot j}$ | 30 | 26 | 25 | 27 | G = 108 |

Here, $t = 3$, $r = 4$, $N = 12$

- For CF

$$CF = \frac{G^2}{N}$$

$$= \frac{108^2}{12}$$

$$= 972$$

- For TSS

$$TSS = \sum\sum y_{ij}^2 - CF$$

$$= 8^2 + 8^2 + \dots 9^2 - 972$$

$$TSS = 26$$

$\bar{x} / P$

<u>For SST</u>

$$SST = \frac{\sum y_{i.}^2}{r} - CF$$

$$= \frac{30^2}{4} + \frac{37^2}{4} + \frac{41^2}{4} - 972$$

$SST = 15.5$

<u>For SSB</u>

$$SSB = \frac{\sum y_{.j}^2}{t} - CF$$

$$= \frac{30^2}{3} + \frac{26^2}{3} + \frac{25^2}{3} + \frac{27^2}{3} - 972$$

$= \cancel{300.7}\ 4.67$

<u>For SSE</u>

$SSE = TSS - SST - SSB$

$= 26 - 15.5 - 4.67$

$\therefore SSE = 5.83$

ANOVA table for RBD

| SV | df | S.S | MSS | F-ratio |
|---|---|---|---|---|
| Treatment | 2 | 15.5 | 7.75 | $F_T = 7.99$ |
| Block | 3 | 4.67 | 1.56 | $F_B = 1.61$ |
| Error | 6 | 5.83 | 0.97 | |
| Total | $12-1=11$ | 26 | | |

* For treatments

Here, we setup hypothesis as

$H_0: \mu_A = \mu_B = \mu_C$ i.e There is no sig. difference in treatments

v/s $H_1: \mu_i \neq \mu_j$ (for atleast one pair) i.e There is sig. difference in at least one pair of treatments

Under $H_0$, test statistics

$$F = \frac{MSST}{MSSE}$$

$F_T = 7.99$

Here,

level of sig ($\alpha$) = 5% (let)

d.f = (2, 6)

From table,

$F_{tab} = 5.143$

Decision:-

Since $F_{cal} > F_{tab}$ we reject $H_0$ & accept $H_1$ with the conclusion that there is sig. difference in at least one pair of treatments

* For Blocks

Here we setup hypothesis as

$H_0: \mu_I = \quad = \mu_{III}$ ie There is no sig. difference in Blocks v/s $H_1: \mu_i \neq \mu_j$ (For atleast one pair) i.e. There is sig. difference in at least one pair of blocks.

Under $H_0$, test statistics

$$F = \frac{MSSB}{MSSE}$$

$F_B = 1.61$

Here, level of sig. $(\alpha) = 5\%$ (let)

d.f $= (3, 6)$

From table,

$F_{tab} = 4.757$

- Decision:-

Since $F_{cal} < f_{tab}$ we accept $H_0$ & reject $H_1$ with the conclusion there there is no sig. difference in blocks.

* Relative efficiency w.r.t CRD

$$E = \frac{r(t-1)\,MSE + (r-1)\,MSB}{(rt-1)\,MSE}$$

$$= \frac{4 \times 2 \times 0.97 + 3 \times 1.56}{(12-1) \times 0.97}$$

$\therefore E = 1.167$

since, $E>1$ so, given design is 16.7% more efficient than CRD

Estimation of missing value in RBD

→ let us consider RBD with 't' treatments & 'r' blocks whose one obs. of ith treatment & jth block has been missing.

$$\text{let } y_{ij} = x$$

Here we estimate '$x$' in such a way that sum of square due to Error will be minimised.

Here,

| | |
|---|---|
| $CF = \frac{G^2}{N}$ | $y_{..}'$ = sum of all known obs. |
| $\therefore CF = \frac{(y_{..}' + x)^2}{rt}$ | $y_{i.}'$ = sum of all known obs. of ith treatment |
| | $y_{.j}'$ = Sum of all known obs of jth block |

For TSS

$$TSS = \sum\sum y_{ij}^2 - C.F$$

$$= x^2 + \text{constant} - \frac{(y_{..}' + x)^2}{r}$$

For SST

$$SST = \frac{\sum y_{i.}^2}{r} - CF$$

$$= \frac{(y_{i.}'+x)^2}{r} + \text{constant} - \frac{(y_{..}'+x)^2}{rt}$$

For SSB

$$SSB = \frac{\sum y_{.j}^2}{t} - CF$$

$$SSB = \frac{(y_{.j}'+x)^2}{t} + \text{constant} - \frac{(y_{..}'+x)^2}{rt}$$

Now,

$$SSE = TSS - SST + SSB$$

$$\therefore SSE = x^2 - \frac{(y_{i.}'+x)^2}{r} - \frac{(y_{.j}'+x)^2}{t} + \text{constant} + \frac{(y_{..}'+x)^2}{rt} \quad \text{--- (ii)}$$

Now,

$$\frac{\partial SSE}{\partial x} = 0 \ [\text{for SSE to be min.}]$$

$$\text{or, } 2x - \frac{2(y_{i.}'+x)}{r} - \frac{2(y_{.j}'+x)}{t} + \frac{2(y_{..}'+x)}{rt} = 0$$

$$\text{or, } x.rt - t(y_{i.}'+x) - r(y_{.j}'+x) + (y_{..}'+x) = 0$$

$$\text{or, } x(rt - t - r + 1) = t y_{i.}' + r y_{.j}' - y_{..}'$$

$$\text{or } x(r-1)(t-1) = t y_{i.}' + r y_{.j}' - y_{..}'$$

$$\boxed{\therefore \ x = \frac{t y_{i.}' + r y_{.j}' - y_{..}'}{(t-1)(r-1)}} \quad \text{--- (iii)}$$

eq$^n$ (26) represents the estimating eq$^n$ for missing obs. in RBD.

14. Given, design is RBD with 3 - treatments & 4 blocks, where one obs. is missing

let $y_{ij} = x$

Now, calc. table for various sums

| Treat | Blocks I | II | III | IV | $y_{i.}$ (93.3) |
|---|---|---|---|---|---|
| P | 19 | x( ) | 23 | 26 | 68 + x |
| Q | 26 | 28 | 27 | 33 | 114 |
| R | 21 | 29 | 22 | 26 | 98 |
| $y_{.j}$ | 66 | 57+x (82.3) | 72 | 85 | G = x + 280 = 305.3 |

For estimation of x

$$x = \frac{t y_{i.}' + r y_{.j}' - y_{..}'}{(t-1)(r-1)}$$

Here, $t = 3$, $r = 4$, $y_{i.}' = 68$, $y_{.j}' = 57$, $y_{..}' = 280$

$$x = \frac{3 \times 68 + 4 \times 57 - 280}{2 \times 3}$$

$x = 25.3$

For CF

$$CF = \frac{G^2}{N} = \frac{305.3^2}{12} = 776.34$$

For TSS

$$TSS = \sum\sum y_{ij}^2 - CF$$

$$= 18^2 + 25.3^2 + \dots + 26^2 - 7767.34$$

$$TSS = 158.75$$

For SST

$$SST = \frac{\sum y_{i.}^2}{r} - CF$$

$$= \frac{93.3^2}{4} + \frac{114^2}{4} + \frac{98^2}{4} - 7767.34$$

$$= 58.88$$

For SSB

$$SSB = \frac{\sum y_{.j}^2}{t} - CF$$

$$= \frac{66^2}{3} + \frac{82.3^2}{3} + \frac{72^2}{3} + \frac{85^2}{3} - 7767.34$$

$$\therefore SSB = 78.76$$

For adjustment Factor

$$AF = \frac{(y_{.j}' + t y_{i.}' - y_{..}')^2}{t(t-1)(r-1)2}$$

$$= \frac{(57 + 3 \times 68 - 280)^2}{3 \times 2 \times 3^2}$$

$= 6.68$

Now,

$SSTA = 58.88 - 6.68$

$= 52.2$

For SSE

$SSE = TSS - SSB - SSTA$

$= 158.75 - 78.75 - 52.2$

$\therefore SSE = 27.79$

Now,

ANOVA table for RBD

| SV | df | SS | MSS | F-ratio |
|---|---|---|---|---|
| treat. | $3-1=2$ | 52.2 | 26.1 | $F_T = 4.69$ |
| block | $4-1=3$ | 78.75 | 26.25 | $F_B = 4.72$ |
| Error | 5 | 27.8 | 5.56 | |
| Total | $12-2=10$ | | | |

* For treatments

Here, we setup hypothesis as,

## latin Square Design

* → It is the design based on all three basic principle of exp. design viz. randomization, Replication & local control. Here treatments are allocated in experimental unit in such a way that each treatment must occur once and only once in each row and column. This design is also known as treatment wise, row wise, column wise design. Here as well treatments are replicated in fair manner inorder to increase precision of a design and to reduce Errom sum of square. Since latin alphabets are used to denote treatments, it is also known as latin square ~~treatm~~ ~~ent~~ Design. We use incomplete three way ANOVA to analyze the LSD.

### Layout of LSD

→ Let us consider 4x4 LSD then layout of LSD maybe

| | | | |
|---|---|---|---|
| B | A | D | C |
| A | ~~B~~C | B | D |
| D | B | C | A |
| C | D | A | B |

### Layout of SLSD

let us consider 4x4 LSD, then layout of SLSD is

| | | | |
|---|---|---|---|
| A | B | C | D |
| B | C | D | A |
| C | D | A | B |
| D | A | B | C |

Statistical model of LSD

→ let us consider $m \times m$ LSD, then statistical model of LSD is

$$y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ijk} \quad \text{— (i)}$$

Here,
$i = 1, 2, \ldots, m$
$j = 1, 2, \ldots, m$
$k = 1, 2, \ldots, m$

where

$y_{ijk}$ = yield or obs. of $i^{th}$ row, $j^{th}$ column & $k^{th}$ treatment

$\mu$ = overall mean

$\alpha_i$ = $i^{th}$ row effect

$\beta_j$ = $j^{th}$ column effect

$\tau_k$ = $k^{th}$ treatment effect

$\epsilon_{ijk}$ =

Statistical ~~An~~ Analysis of LSD

For the analysis of LSD, we partition total sum of square into its component parts as,

$$\sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(y_{ijk}-\bar{y}_{...})^2 = \sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(y_{ijk}-\bar{y}_{i..}-\bar{y}_{.j.}-\bar{y}_{..k}+\bar{y}_{i..}+\bar{y}_{.j.}+\bar{y}_{..k}-\bar{y}_{...})^2$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}\{(\bar{y}_{i..}-\bar{y}_{...})+(\bar{y}_{.j.}-\bar{y}_{...})+(\bar{y}_{..k}-\bar{y}_{...}) + (y_{ijk}-\bar{y}_{i..}-\bar{y}_{.j.}-\bar{y}_{..k}+2\bar{y}_{...})\}^2$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(\bar{y}_{i..}-\bar{y}_{...})^2+\sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(\bar{y}_{.j.}-\bar{y}_{...})^2+$$

$$\sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(\bar{y}_{..k}-\bar{y}_{...})^2+\sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(y_{ijk}-\bar{y}_{i..}+$$

$$\bar{y}_{..k}+2\bar{y}_{...})^2 + \text{Cross product term}$$

$$= m\sum_{i=1}^{m}(\bar{y}_{i..}-\bar{y}_{...})^2+m\sum_{j=1}^{m}(\bar{y}_{.j.}-\bar{y}_{...})^2$$

$$+m\sum_{k=1}^{m}(\bar{y}_{..k}-\bar{y}_{...})^2+\sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m}(y_{ijk}-\bar{y}_{i..}-\bar{y}_{.j.}-$$

$$\bar{y}_{..k}+2\bar{y}_{..})^2$$ [Cross product product term vanishes as sum of deviation taken from A.M is zero]

$\therefore$ $\boxed{TSS = SSR + SSC + SST + SSE}$

For practical purpose

$$TSS = \sum_{i=1}^{m}\sum_{j=1}^{m}\sum_{k=1}^{m} y_{ijk}^2 - CF$$

$$SSE = TSS - SSR - SSC - SST$$

$$CF = \frac{G^2}{N}, \; N = m^2$$

$$SSR = \frac{\sum_{i=1}^{m} y_{i..}^2}{m} - CF.$$

$$SSC = \frac{\sum_{j=1}^{m} y_{.j.}^2}{m} - CF$$

$$SST = \frac{\sum_{k=1}^{m} y_{..j}^2}{m} - CF$$

ANOVA table for LSD.

| SV | df | SS | MSS | F |
|---|---|---|---|---|
| Rows. | $m-1$ | SSR | $MSSR = \frac{SSR}{m-1}$ | $F_R$ = M[illegible] |
| column | $m-1$ | SSC | $MSSC = \frac{SSC}{m-1}$ | $F_C$ = M[illegible] |
| Treat. | $m-1$ | SST | $MSST = \frac{SST}{m-1}$ | $F_T$ = M[illegible] |
| Error | $(m-1)(m-2)$ | SSE | $MSSE = \frac{SSE}{(m-1)(m-2)}$ | |
| Total | $m^2-1$ | TSS | | |

Decision:-

* If $F_{cal.} \le F_{tab}$, we acept $H_0$ & reject $H_1$

* If $F_{cal.} > F_{tab}$, we reject $H_0$ & acept $H_1$

13.)

→ soln:-

Given design is 3×3 LSD.

Now, calculation table for various sum.

| Rows | Column I | Column II | Column III | $y_{i..}$ |
|---|---|---|---|---|
| I | 10 | 15 | 20 | 45 |
| II | 25 | 10 | 15 | 50 |
| III | 25 | 20 | 15 | 60 |
| $y_{.j.}$ | 60 | 45 | 50 | 155 |

m = 3

For treatment,

| | $y_{..k}$ |
|---|---|
| A: 10+20+15 | 45 |
| B: 25+15+15 | 55 |
| C: 25+10+20 | 55 |
| | 155 |

For C.F

$$CF = \frac{G^2}{N}$$

$$= \frac{155^2}{9}$$

$\therefore CF = 2669.44$

For TSS

$$TSS = \sum\sum\sum y_{ijk}^2 - CF$$

$$= 10^2 + 15^2 + \ldots + 15^2 - 2669.44$$

$\therefore TSS = 255.56$

For SSR

$$SSR = \frac{\sum y_{i..}^2}{m} - CF$$

$$= \frac{45^2}{3} + \frac{50^2}{3} + \frac{60^2}{3} - 2669.44$$

$\therefore SSR = 38.89$

For SSC

$$SSC = \frac{\sum y_{.j.}^2}{m} - CF$$

$$= \frac{60^2 + 45^2 + 50^2}{3} - 2669.44$$

$\therefore SSC = 38.89$

For SST

$$SST = \frac{\sum y_{\cdot \cdot k}^2}{m} - CF$$

$$= \frac{45^2 + 55^2 + 55^2}{3} - 2669.44$$

$\therefore SST = 22.23$

For SSE

$SSE = TSS - SSR - SSC - SST$

$= 255.56 - 38.89 - 38.89 - 22.23$

$SSE = 155.55$

ANOVA table for LSD

| SV | df | SS | MSS | F-ratio |
|---|---|---|---|---|
| Row | 2 | 38.89 | 19.445 | $F_R = 0.25$ |
| Column | 2 | 38.89 | 19.445 | $F_C = 0.25$ |
| Treat- | 2 | 22.23 | 11.115 | $F_T = 0.14$ |
| Error | 2 | 155.55 | 77.775 | |
| Total | 8 | 255.56 | | |

* For Row

Here we setup hypothesis as

$H_0: \mu_I = \mu_{II} = \mu_{III}$ i.e There is no significant difference due to Row.

v/s $H_1: \mu_i \neq \mu_j$ (for atleast one pair) i.e There is significant difference due to atleast one pair.

Under $H_0$, test statistics

$F_R = \frac{MSSR}{MSSE} \Rightarrow F_R = 0.25$

Here,

level of significance $(\alpha) = 5\%$

$d.f = (2,2)$

From table,

$F_{tab} = 19$

Decision:- Since, $F_{cal} < F_{tab}$ we acept $H_0$ & reject $H_1$ with the conclusion that there is no significant difference due to row

2nd part

(i) Relative efficiency w.r.t RBD

$$E = \frac{(m-1)\,MSE + MSC}{m \cdot MSE}$$

$$= \frac{(3-1) \times 77.775 + \cancel{28.89}\ 19.445}{3 \times 77.775}$$

$$= \cancel{0.833}\ 0.75$$

since $E < 1$ so, given design is less efficient than RBD

when column is treated as block

$$E = \frac{(m-1)\,MSE + MSR}{m\,MSE}$$

$$= \frac{(3-1) \times 77.775 + 19.445}{3 \times 77.775}$$

$= 0.75$

since $E < 1$, so, the given design is 25% less efficient than RBD

(ii) Relative efficiency wrt CRBD

$$E = \frac{(m-1) MSE + MSC + MSR}{(m+1) MSE}$$

$$= \frac{2 \times 77.775 + 19.445 + 19.445}{4 \times 77.775}$$

$$= 0.625$$

since $E < 1$, so given data is 37.5% less efficient then CRD.

* Estimation of missing value in LSD

letus consider $m \times m$ LSD in which one observation from $i^{th}$ row, $j^{th}$ column & $k^{th}$ treatment is missing.

Let $y_{ijk} = x$

Here we estimate $x$ in a such way that sum of square due to error will be minimized.

For CF

$$CF = \frac{(y'_{...} + x)^2}{m^2}$$

* For TSS

$$TSS = \sum\sum\sum y_{ijk}^2 - CF$$

$$= x^2 + constant - \frac{(y_{...}' + x)^2}{m^2}$$

* For SSR

$$SSR = \sum \frac{y_{i..}^2}{m} - CF$$

$$= \frac{(y_{i..}' + x)^2}{m} + constant - \frac{(y_{...}' + x)^2}{m^2}$$

* For SSC

$$SSC = \sum \frac{y_{.j.}^2}{m} - CF$$

$$= \frac{(y_{.j.} + x)^2}{m} + const - \frac{(y_{...}' + x)^2}{m^2}$$

* SST

$$SST = \sum \frac{y_{..k}^2}{m} - CF$$

$$= \frac{(y_{..k}' + x)^2}{m} + const. - \frac{(y_{...}' + x)^2}{m^2}$$

Now,

$$\therefore SSE = TSS - SSR - SSR - SSC - SST$$

$$= x^2 - \frac{(y_{i..}' + x)^2}{m} - \frac{(y_{.j.}^2 + m)^2}{m} - \frac{(y_{..k}' + m)^2}{m}$$

$$+ const. + 2\frac{(y_{...}' + x)^2}{m^2}$$

$$\frac{dSSE}{dx} = 0$$

or $$2x - \frac{2(y_{i..}'+x)}{m} - \frac{2(y_{.j.}'+x)}{m} + \frac{2(y_{..k}'+x)}{m}$$

$$+ \frac{4(y_{...}'+m)}{m^2} = 0$$

or $$m^2x - m(y_{i..}'+x) - m(y_{.j.}'+x) + m(y_{..k}'+x)$$
$$+ 2(y_{...}'+x) = 0$$

or $$(m^2 - 3m + 2)\,x = m(y_{i..}' + y_{.j.}' + y_{..k}') - 2y_{...}'$$

or $$(m-1)(m-2)\,x = m(y_{i..}' + y_{.j.}' + y_{..k}') - 2y_{...}'$$

$$\boxed{x = \frac{m(y_{i..}' + y_{.j.}' + y_{..k}') - 2y_{...}'}{(m-1)\quad(m-2)}} \quad (22)$$

where,

$y_{i..}'$ = Sum of all known obs. of $i^{th}$ row
$y_{.j.}'$ = Sum of all known obs. of $j^{th}$ column
$y_{..k}'$ = Sum of all known obs. of $k^{th}$ treatment
$y_{...}'$ = Sum of all known obs.

Eqn (22) represents estimating expression of missing obs. in L.SD.

15) Soln:
Given design is 4x4 LSD in which one observation is missing.
let missing obs. be $x$.

Now,
calcn table for various Sums

Column

| Row | I | II | III | IV | $y_{i..}$ |
|---|---|---|---|---|---|
| I | 12 | 19 | 10 | 8 | 49 |
| II | 18 | 12 | 6 | $x$ (2) | $36+x$ (38) |
| III | 22 | 10 | 5 | 21 | 58 |
| IV | 12 | 7 | 27 | 17 | 63 |
| $y_{.j.}$ | 64 | 48 | 48 | $x+46$ (48) | $G = x+206$ (208) |

For treatment,

| | $y_{..k}$ |
|---|---|
| A : $12+7+5+x$ | $x+24$ : 26 |
| B : $22+12+10+17$ | 61 |
| C : $18+19+27+21$ | 85 |
| D : $12+10+6+8$ | 36 |
| | $G = x+206$ |

For estimation of missing value

$$x = \frac{m(y_{i..}' + y_{.j.}' + y_{..k}') - 2y_{...}'}{(m-1)(m-2)}$$

$$= \frac{4(36+46+24) - 2\times 206}{3\times 2}$$

$\therefore x = 2$

Now, various sums

$$CF = \frac{G^2}{N}$$

$$= \frac{(208)^2}{16}$$

$$= 2704$$

For TSS

$$TSS = \sum\sum\sum y_{ijk}^2 - CF$$

$$= 12^2 + 19^2 + \dots + 17^2 - 2704$$

$$TSS = 734$$

For SSR

$$SSR = \frac{\sum y_{i..}^2}{m} - CF$$

$$= \frac{49^2 + 38^2 + 58^2 + 63^2}{4} - 2704$$

$$SSR = 90.5$$

For SSC

$$SSC = \frac{\sum y_{.j.}^2}{m} - CF$$

$$= \frac{64^2 + 48^2 + 48^2 + 48^2}{4} - 2704$$

$$= 48$$

For SST

$$SST = \frac{\sum y_{..k}^2}{m} - CF$$

$$= \frac{26^2 + 61^2 + 85^2 + 66^2}{4} - 2704$$

$$SST = 525.5$$

Now

Adj factor :-

$$AF = \frac{\{(m-1)\, y_{..k}' + y_{i..}' + y_{.j.}' - y_{...}'\}^2}{\{(m-1)(m-2)\}^2}$$

$$= \frac{\{3 \times 24 + 36 + 48 - 206\}^2}{\{3 \times 2\}^2}$$

$$A.F = 75.11$$

Now SSTA

$$SSTA = 525.5 - 75.11$$

$$= 450.39$$

For SSE

$$SSE = TSS - SSR - SSC - SSTA$$

$$= 734 - 90.5 - 48 - 450.39$$

$$= 145.11$$

Noco
ANOVA table for LSD

| SV | df | SS | MSS | F-ratio |
|---|---|---|---|---|
| Row | 4-1=3 | 90.5 | 30.17 | $F_R = 1.04$ |
| column | 3 | 48 | 16 | $F_C = 0.55$ |
| Treatment | 3 | 450.39 | 150.13 | $F_T = 5.17$ |
| error | 5 | 145.11 | 29.02 | |
| Total | 16-2=14 | 734 | | |

For Row,

Here we setup hypothesis as

$H_0: \mu_I = \mu_{II} = \dots = \mu_{IV}$ ie there is no significant difference due to row

V/s $H_1: \mu_i \neq \mu_j$ ie there is significant difference due to ~~row~~ atleast one pair

Under $H_0$, test statistics,

$F_R = 1.04$

Here

level of sig ($\alpha$) = 0.05

df = (3,5)

From table,

$F_{tab} = 5.409$

Decision:- Since $F_{cal} < F_{tab}$ we accept $H_0$ & reject $H_1$ with the conclusion that there is no significant difference due to row.

For treatment:

Here we setup hypothesis as:-

$H_0: \mu_I = \mu_{II} = \ldots = \mu_{IV}$ ie there is no significant differences due to treatment

v/s

$H_1: \mu_i \neq \mu_j$ ie there is significant difference due to ~~treat~~ at least one pair of treatment

Under $H_0$, test statistics,

$$F_t = 5.17$$

Here

level of significance $(\alpha) = 0.05$

$df = (3, 5)$

From table,

$$F_{tab.} = 5.409$$

Decision:- Since $F_{cal} < F_{tab.}$ we accept $H_0$ & reject $H_1$ with the conclusion there is no significant differences due to treatment.

16) Here we have

| S.V | d.f | SS | MSS | F-ratio |
|---|---|---|---|---|
| Blocks | 4 | 26.8 | 6.7 | $F_B = 2.68$ |
| Treat. | 3 | 28.5 | 9.5 | $F_T = 3.8$ |
| Error | 12 | 30 | 2.5 | |
| Total | 19 | 85.3 | | |

We know,

$$df \text{ for Blocks} = r-1$$
$$4 = r-1$$
$$r = 5$$

$$df \text{ for treat} = t-1$$
$$3 = t-1$$
$$t = 4$$

$$\text{Error } df = (t-1)(r-1)$$
$$= 4 \times 3$$
$$= 12$$

$$\text{Total } df = t \cdot r - 1$$
$$= \cancel{4\times 5-1}\ 19$$

Again,

$$MSSE = \frac{SSE}{df}$$

$$2.5 \times 12 = SSE$$
$$SSE = 30$$

$$SSB + SST + SSE = TSS$$
$$26.8 + SST + 30 = 85.3$$
$$SST = 28.5$$

Now,

$$MSSB = \frac{SSB}{df}$$

$$= \frac{268}{4}$$

$$= 6.7$$

$$MSST = \frac{SST}{df}$$

$$= \frac{28.5}{3}$$

$$= 9.5$$

Now,

$$F_B = \frac{MSSB}{MSSE}$$

$$= \frac{6.7}{2.5}$$

$$F_B = 2.68$$

$$F_T = \frac{MSST}{MSSE}$$

$$= \frac{9.5}{2.5}$$

$$= 3.8$$

16) → Here we have

| SV | d.f | SS | MSS | F-ratio |
|---|---|---|---|---|
| Column | 5 | 10.3 | 2.06 | 3.17 |
| Row | 5 | 4.2 | 0.84 | 1.29 |
| Treatm. | 5 | 12.15 | 2.43 | 3.74 |
| Error | 20 | 13 | 0.65 | |
| Total | 35 | 39.65 | | |

We Know,

$$df \text{ of columns} = m-1$$

$$5 = m-1$$

$$m = 6$$

$$df \text{ of rows} = df \text{ of columns}$$
$$= m-1$$
$$= 6-1$$
$$= 5$$

$$df \text{ for Error} = (m-1)(m-2)$$
$$= (6-1)(6-2)$$
$$= 5\times4$$
$$= 20$$

$$df \text{ for treat} = m-1$$
$$= 6-1$$
$$= 5$$

Again

$$MSST = \frac{SST}{df}, \quad MSSE = \frac{SSE}{df}$$

$$5\times2.43 = SST \qquad 0.65\times20 = SSE$$

$$= 12.15 \qquad SSE = 13$$

$TSS = SSR + SSC + SST + SSE$

$39.65 = 4.2 + 12.15 + 13 + SSC$

$SSC = 10.3$

Now,

$MSSC = \frac{SSC}{df}$ , $MSSR = \frac{SSR}{df}$

$= \frac{10.3}{5}$ $= \frac{4.2}{5}$

$= 2.06$ $= 0.84$

Now,

$F_C = \frac{MSSC}{MSSE}$ , $F_R = \frac{MSSR}{MSSE}$ , $F_T = \frac{MSST}{MSSE}$

$= \frac{2.06}{0.65}$ $= \frac{0.84}{0.65}$ $= \frac{2.43}{0.65}$

$= 3.17$ $= 1.29$ $= 3.74$

* Stochastic process

→ A family of random variables indexed by a parameter (time) it is called stochastic process. It is denoted as $X_t$ (Stochastic Variable)

* Markov process:

A stochastic process in which future state only depends only on the present state is called Markov process.

* Transction Probability

The probability of moving from one state to another or remain in the same state in a single period of time is called transction probability matrix.

It is a conditional probability.

Here

$$P\{\underbrace{x_{(t+1)}}_{\text{step}} = j / x_{(t)} = i\} = p_{ij}(1)$$

→ state

where

$p_{ij}(1)$ denotes transition probability of changing state i to j

Transition Probability matrix:

It is a matrix obtained by transition probability of various states.

Let, $I = \{0, 1, 2, 3, \ldots, m\}$ be state space
then transition probability matrix is

$$P = \begin{bmatrix} P_{00} & P_{01} & \cdots & P_{0n} \\ P_{10} & P_{11} & \cdots & P_{1n} \\ \vdots & \vdots & & \vdots \\ P_{n0} & P_{n1} & \cdots & P_{nn} \end{bmatrix}$$

where, $P_{ij} \geq 0$
$\& \sum_{j} P_{ij} = 1$

Q8)
Here we have
Transition prob matrix

$$P = \begin{bmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{bmatrix}$$

i) Here, two step transition prob matrix

$$P^2 = \begin{pmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{pmatrix}$$

$$= \begin{pmatrix} 0.37 & 0.63 \\ 0.36 & 0.64 \end{pmatrix}$$

ii) Here,
Reqd prob $= P_{11}^{(3)}$

$= P_{11} P_{11} P_{11} + P_{11} P_{12} P_{21} + P_{21} P_{22} P_{21} + P_{12} P_{21} P_{11}$

$= 0.027 + 0.084 + 0.168 + 0.084$
$= 0.363$

$$\begin{pmatrix} 0.37 & 0.63 \\ 0.36 & 0.64 \end{pmatrix}\begin{pmatrix} 0.3 & 0.7 \\ 0.4 & 0.6 \end{pmatrix}$$

* Steady state prob:- It is the probability of a state in long run. It is generally denoted by ($\pi$)

Such $\pi p = \pi$

where
$p$ = Transition prob. matrix

Here,
$\pi = \pi_0, \pi_1, \pi_2, \ldots, \pi_n)$
$\Sigma x_p = 1$

9.)
→ soln:-
let short be 1st state
Tall be 2nd state

(i) By question, TPM,

$$p = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix}$$

$$\therefore p = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

(ii) Req$^d$ prob. $= P_{21}(2)$
$= P_{22} P_{21} + P_{21} P_{11}$
$= 0.6 \times 0.4 + 0.4 \times 0.7$
$= 0.52$

10)

→ soln:

(i) Here we have

$$\begin{bmatrix} 0.4 & 0.6 & 0 \\ 0 & 0.7 & 0.3 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

We know, $\sum_{j} P_{ij} = 1$

(ii) let $\pi = (\pi_1 \ \pi_2 \ \pi_3)$ be steady state prob., then

$$\pi P = \pi$$

or, $$(\pi_1 \ \pi_2 \ \pi_3)\begin{pmatrix} 0.4 & 0.6 & 0 \\ 0 & 0.7 & 0.3 \\ 0.2 & 0.2 & 0.6 \end{pmatrix} = (\pi_1 \ \pi_2 \ \pi_3)$$

$$(0.4\pi_1 + 0.2\pi_3 \quad 0.6\pi_1 + 0.7\pi_2 + 0.2\pi_3 \quad 0.3\pi_2 + 0.6\pi_3) = (\pi_1 \quad \pi_2 \quad \pi_3)$$

By defn of equal matrices

or, $0.4\pi_1 + 0.2\pi_3 = \pi_1$

or, $0.6\pi_1 = 0.3\pi_3 \Rightarrow \pi_3 = 3\pi_1$ — (i)

&

or, $0.3\pi_2 + 0.6\pi_3 = \pi_3$

or, $0.3\pi_2 = 0.4\pi_3$

$3\pi_2 = 4 \times 3\pi_1$ (using eqn (i))

$\pi_2 = 4\pi_1$

We know,

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_1 + 4\pi_1 + 3\pi_1 = 1$$

$$8\pi_1 = 1$$

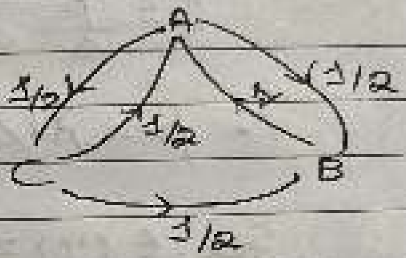
$$\therefore \pi_1 = 1/8$$

11) Soln:-

Here by question,

TPM for given information

$$P = \begin{array}{c c} & \begin{array}{ccc} A & B & C \end{array} \\ \begin{array}{c} A \\ B \\ C \end{array} & \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1 & 0 & 0 \\ 1/2 & 1/2 & 0 \end{pmatrix} \end{array}$$




$$\pi P = P$$

where $\pi = (\pi_0 \; \pi_1 \; \pi_2)$

$$(\pi_1 \; \pi_2 \; \pi_3) \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1 & 0 & 0 \\ 1/2 & 1/2 & 0 \end{pmatrix} = (\pi_1 \; \pi_2 \; \pi_3)$$

$$= \pi_2 + \pi_3/2 \cdot 0.5\pi_1 + 0.5\pi_3$$

$$\cdot \; 0.5\,\pi_1$$

By defn of equal matrices

| | |
|---|---|
| $\pi_2 + 0.5\pi_3 = \pi_1$ | $\pi_2 + 0.5\pi_3 = 2\pi_3$ |
| $0.5\pi_1 + 0.5\pi_3 = \pi_2$ | $\pi_2 = 1.5\pi_3$ |
| $0.5\pi_1 = \pi_3$ | |
| $\pi_1 = \dfrac{\pi_3}{0.5}$ | $\pi_1 + \pi_2 + \pi_3 = 1$ |
| | $2\pi_3 + 1.5\pi_3 + \pi_3 = 1$ |
| $\pi_1 = 2\pi_3$ | $4.5\pi_3 = 1, \; \pi_3 = 1/4.5$ |

Binomial Counting Process:- It is descrete time descrete space counting process.

Notation

$\lambda$ = Arrival rate
$\Delta$ = Frame size
$p$ = prob. of success
$T$ = Inter arrival time

Computational Formula

$$\lambda = \frac{p}{\Delta}, \quad n = t/\Delta$$

$$\therefore p = \lambda \cdot \Delta, \quad T = Y\Delta$$

$$\text{So, } E(T) = \frac{\pi}{\lambda}$$

$$V(T) = \frac{1-p}{\lambda^2}$$ //

13) let, $X$ = No. of tasks

Arrival rate $(\lambda) = 2$ tasks per minute.
Frame size $(\Delta) = 4$ sec. Frames $= \frac{4}{60} = \frac{1}{15}$ min. Frames

we know, $p = \lambda \cdot \Delta = 2 \times \frac{1}{15} = \frac{2}{15}$

i) For $t = 8$

$$n = t/\Delta = \frac{8}{1/15}$$

Now, $n = 120$

$\therefore$ Reqd. prob $= p(x_{(t)} > 3)$

$= 1 - [p(x_{(t)} = 0) + p(x_{(t)} = 1) + p(x_{(t)} = 2) + p(x_{(t)} = 3)]$

$= 1 - [{}^nC_0\, p^0 q^{n-0} + {}^nC_1\, p^1 q^{n-1} + {}^nC_2\, p^2 q^{n-2} + {}^nC_3\, p^3 q^{n-3}]$

$= 1 - \left[ \left(\frac{13}{15}\right)^{120} + {}^{120}C_1 \times \frac{2}{15} \times \left(\frac{13}{15}\right)^{119} + {}^{12}C_2 \left(\frac{2}{15}\right)^2 \times \left(\frac{13}{15}\right)^{118} + {}^{12}C_3 \left(\frac{2}{15}\right)^3 \left(\frac{13}{15}\right)^{117} \right]$

$= 0.999$

(b) Here $t = 50$ mins

$n = t/\Delta = \frac{50}{1/15} = 750$

For Normal approximation

$\mu = np = 750 \times \frac{2}{15} = 100$

$\sigma^2 = npq = 750 \times \frac{2}{15} \times \frac{13}{15}$

$\sigma = 9.31$

Now,

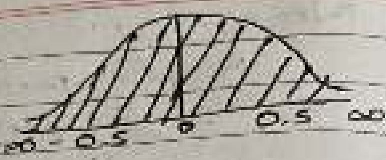
reqd prob. $= p(x_{(t)} > 10)$

$= p(x_{(t)} > 10.5)$ (Using continuity correction factor)

$= p\left(z > \frac{10.5 - 100}{9.31}\right)$ $\quad z = \left(\frac{x_t - \mu}{\sigma}\right)$

$= p(z > -9.61)$

$\approx 1$




104) Here we have,

arrival rate $(\lambda) = 3$ per min

i) Here we have,

$\rho = 0.2$, $s = ?$

We know,

$$\rho = \lambda \cdot s$$

$$\frac{0.2}{3} = s$$

$$s = 1/15 \ \#$$

(ii) For $E(T)$

We know,

$$1/\lambda = 1/3$$

For $V(T)$

$$V(T) = \frac{1-\rho}{\lambda^2}$$

$$= \frac{1-0.2}{3^2}$$

$$= \frac{0.8}{9}$$

$$\Rightarrow S.D(T) = \sqrt{\frac{0.8}{9}} = 0.298$$

* Poisson Process:-

It is the continuous time counting process & is limiting case of binomial process.

Notations:-

$\lambda$ = arrival ~~time~~ rate

$t$ = time period

$T$ = inter arrival time

Here,

$E(T) = \lambda t$

& $V(T) = \lambda t$

19)

Here we have,

$\lambda = 2$ calls in 5 mins

$\therefore \lambda = \frac{2}{5}$ calls in 1 min

For $t = 15$

$\lambda t = 15 \times \frac{2}{5} = 6$

Now,

Req. prob. $= p(x > 10)$

$= 1 - [p(x=10) + p(x=9) + \cdots + p(x=0)]$

$= 1 - e^{-\lambda t}\left[\frac{(\lambda t)^{10}}{10!} + \frac{(\lambda t)^{9}}{9!} + \cdots + \frac{(\lambda t)^{0}}{0!}\right]$

$= 1 - e^{-6}\left[\frac{6^{10}}{10!} + \frac{6^{9}}{9!} + \frac{6^{8}}{8!} + \frac{6^{7}}{7!} + \cdots + 6 + 1\right]$

$= 0.043$

## Bernoulli Single Server Queing System

→ It is discrete time queuing process with one server and unlimited capacity.

Here,

$P_A = \lambda\Delta$

$P_S = \mu\Delta$

$P_{00} = P(\text{no arrivals}) = 1 - P_A$

$P_{01} = P(\text{one arrival}) = P_A$

For all $i \geq 1$

~~$P_{i,i-1} = P(\text{one departure})$~~

$P_{i,i-1} = P(\text{no arrival} \cap \text{one departure})$
$= (1 - P_A) . P_S$

$P_{i,i} = P(\text{no arrival} \cap \text{no departure})$ or
$P(\text{one arrival} \cap \text{one departure})$
$= (1 - P_A)(1 - P_S) + P_A P_S$

$P_{i,i+1} = P(\text{one arrival} \cap \text{no departure})$
$= P_A \cdot (1 - P_S)$

Now,

TPM,

$$P = \begin{bmatrix} 1 - P_A & P_A & 0 & 0 & 0 \\ (1-P_A)P_S & (1-P_A)(1-P_S)+P_AP_S & P_A(1-P_S) & 0 & \\ 0 & (1-P_A)P_S & (1-P_A)(1-P_S)+P_AP_S & P_A(1-P_S) & \end{bmatrix}$$

2a)

Soln:-

Here we have,

Arrival rate $(\lambda) = 9$ per hr

Service rate $(\mu) = \frac{60}{3} = 20$ per hr

Frame size $(\Delta) = 15$ sec

$= \frac{15}{60 \times 60}$ hr

$\therefore \Delta = \frac{1}{240}$ hr

Now,

$P_A = \lambda \Delta$

$= 9 \times \frac{1}{240}$

$= \frac{3}{80}$

$P_S = \mu \Delta$

$= 20 \times \frac{1}{240}$

$= \frac{1}{12}$

For $i = 0$

$P_{00} = 1 - P_A$

$= 1 - \frac{3}{80}$

$= 77/80$

$P_{01} = P_A$

$= 3/80$

For $i \geq 1$

$$P_{i,i-1} = (1-P_A)P_S$$
$$= \frac{77}{80} \times \frac{1}{12}$$
$$= \frac{77}{960}$$

$$P_{i,i} = (1-P_A)(1-P_S) + P_A P_S$$
$$= \frac{77}{80} \times \frac{11}{12} + \frac{3}{80} \times \frac{1}{12}$$
$$= \frac{850}{960}$$

$$P_{i,i+1} = P_A(1-P_S)$$
$$= \frac{3}{80} \times \frac{11}{12}$$
$$= \frac{33}{960}$$

Now, TPM,

$$P = \begin{bmatrix} 77/80 & 3/80 & 0 & 0 & 0 & \cdots \\ 77/960 & 850/960 & 33/960 & 0 & 0 & \cdots \\ 0 & 77/960 & 850/960 & 33/960 & 0 & \cdots \\ 0 & 0 & 77/960 & 850/960 & 33/960 & \cdots \end{bmatrix}$$

24) Soln:-

Here we have,

Arrival time/rate ($\lambda$) = 8 per hr

Service rate ($\mu$) = $60/3$ = 20 per hr.

frame size ($\Delta$) = 5 sec

$$= \frac{5}{60 \times 60}$$

$$\therefore \Delta = \frac{1}{720} \text{ hr}$$

Now,

$$P_A = \lambda\Delta = 8 \times \frac{1}{720}$$

$$= \frac{1}{90}$$

$$P_S = \mu\Delta$$

$$= 20 \times \frac{1}{720}$$

$$= \frac{1}{36}$$

For $i = 0$

| | |
|---|---|
| $P_{00} = 1 - P_A$ | $P_{01} = P_A$ |
| $= 1 - \frac{1}{90}$ | $= \frac{1}{90}$ |
| $= \frac{89}{90}$ | |

For $i \geq 1$

$$P_{i,i-1} = (1-P_A)\cdot P_S = \frac{89}{90} \times \frac{1}{36} = \frac{89}{3240}$$

$$P_{i,i} = (1-P_A)(1-P_S) + P_A P_S$$

$$= \frac{89 \times 35}{90 \quad 36} + \frac{1}{90} \times \frac{1}{36}$$

$$= \frac{3116}{3240}$$

$$P_{i,i+1} = P_A(1-P_S)$$

$$= \frac{1}{90} \times \left(\frac{35}{36}\right)$$

$$= \frac{35}{3240}$$

Now, TPM,

$$P = \begin{pmatrix} 89/90 & 1/90 & 0 & 0 \\ 89/3240 & 3116/3240 & 35/3240 & 0 \\ 0 & 89/3240 & 3116/3240 & 35/3240 \\ 0 & 0 & 89/3240 & 3151/3240 \end{pmatrix}$$

25. → Soln:-

Here we have,

Arrival rate ($\lambda$) = 10 call per hr

Service rate ($\mu$) = $\frac{60}{4}$ = 15 per hr

Frame size ($\Delta$) = 1 min

$\therefore \Delta = \frac{1}{60}$ hr

Now,

$P_A = \lambda\Delta$

$= 10 \times \frac{1}{60}$

$= \frac{1}{6}$

$P_S = \mu\Delta$

$= \cancel{20}\,15 \times \frac{1}{60}$

$= \frac{1}{4}$

For $i = 0$

$P_{00} = 1 - P_A$

$= 1 - \frac{1}{6}$

$= \frac{5}{6}$

$P_{01} = P_A$

$= \frac{1}{6}$

M/M/1 system:- It is continuous time Markov process with 1 server & unlimited capacity.

Computational Formulae

Utilization rate $= \frac{\lambda}{\mu} = \rho$

Idle rate $= 1 - \rho$

Prob. of no customer in queue $= 1-\rho$
" " one " " " $= \rho(1-\rho)$
" " two " " " $= \rho^2(1-\rho)$
" " n " " " $= \rho^n(1-\rho)$

Prob. of server in busy $= \rho$

($L_s$) Expected no of customer in the system $= \frac{\rho}{1-\rho}$

($L_q$) Expected no. of customer in the queue $= \frac{\rho^2}{1-\rho}$

($W_s$) Av. waiting time for system $= \frac{\rho}{\lambda(1-\rho)}$ $\left[W_s = \frac{L_s}{\lambda}\right]$

($W_q$) " " " " queue $= \frac{\rho^2}{\lambda(1-\rho)}$ $\left[W_q = \frac{L_q}{\lambda}\right]$

Prob. of k or more customer in the system $= p(n \geq k) = \rho^k$

Prob of less than k customer in the system $= p(n < k) = 1 - \rho^k$

Variance of queue length $= \frac{\rho}{(1-\rho)^2}$

26a) Here we have,

$\lambda = \frac{1}{6} \times 60 = 10$ per hr. $\rho = \frac{\lambda}{\mu} = \frac{10}{15} = \frac{2}{3}$

$\mu = \frac{1}{4} \times 60 = 15$ per hr.

i) Req^d exp. time (ws) $= \frac{\rho}{\lambda(1-\rho)}$

$= \frac{2/3}{10(1-2/3)}$

$= \frac{1}{5} \times 60$

$= 12$ mins

ii) Req^d prob $= \rho = 2/3$

27) Here we have

$\lambda = \frac{1}{5} \times 60 = 12$ per hr

$\mu = \frac{1}{3} \times 60 = 20$ per hr

Here,

$\rho = \frac{\lambda}{\mu}$

$= \frac{12}{20} = \frac{3}{5}$

a) Req^d exp time $(W_s) = \dfrac{\rho}{\lambda(1-\rho)}$

$$= \frac{3/5}{12(1-3/5)}$$

$$= \frac{1}{8} \times 60$$

$$= 7.5 \text{ mins}$$

b) Req^d prob. $= P(n<2)$

$$= 1 - \rho^2$$

$$= 1 - \left(\frac{3}{5}\right)^2$$

$$= 0.64$$

c) Req^d queue length $(L_q) = \dfrac{\rho^2}{(1-\rho)}$

$$= \frac{(3/5)^2}{(1-3/5)}$$

$$= 0.9$$

29)

→ Here we have,

$\lambda = \dfrac{[illegible]}{4} \times 60$

$= 240$ per hr

$\rho = \dfrac{\lambda}{\mu}$

$= \dfrac{240}{360}$

$= \dfrac{2}{3}$

$\mu = \dfrac{1}{10} \times 60 \times 60 = 360$ per hr

a) First part

exp. no. of customer in the system $(L_s) = \frac{\rho}{(1-\rho)}$

$= \frac{2/3}{(1-2/3)}$

$= 2$

Second part

Variance of queue length $= \frac{\rho}{(1-\rho)^2} = \frac{2/3}{(1-2/3)^2} = 6$

SD of queue length $= \sqrt{6} = 2.44$

b) Reqd prob. $= 1-\rho$
$= 1-2/3$
$= 1/3$
$= 0.33$

31.)
Here we have,

$\lambda = \frac{1}{10} \times 60 = 6$ per hr

$\mu = \frac{1}{6} \times 60 = 10$ per hr

Here,
$\rho = \frac{\lambda}{\mu} = \frac{6}{10} = \frac{3}{5}$

i) Av. queue length $(L_q) = \frac{\rho^2}{1-\rho} = \frac{(3/5)^2}{1-3/5} = 0.9$

(ii) Av. time spent in the system $(W_s) = \frac{\rho}{\lambda(1-\rho)}$

$= \frac{3/5}{6(1-3/5)}$

$= \frac{1}{4} \times 60$

$= 15$ mins

(iii) Reqd prob = $p(n=2) = \rho^2(1-\rho)$

$= \left(\frac{3}{5}\right)^2 (1-3/5)$

$= \frac{18}{125}$

30)

Here, we have,

$\lambda = 30$ per hr

$\mu = \frac{1}{1.5} \times 60$

$= 40$ per hr

Here

$\rho = \frac{\lambda}{\mu}$

$= \frac{30}{40}$

$= 3/4$

Now,

i → Req$^d$ prob $= \rho$

$= 3/4$

ii → Req$^d$ waiting time for system ($W_s$) $= \dfrac{\rho}{\lambda(1-\rho)}$

$= \dfrac{3/4}{30(1-3/4)}$

$= \dfrac{1}{10} \times 60$

$= 6$